

Spectral Estimation Overview

- Periodogram
- Bias, variance, and distribution
- Blackman-Tukey Method
- Welch-Bartlett Method
- Others

Introduction

$$R_x(e^{j\omega}) \triangleq \sum_{l=-\infty}^{\infty} r_x(l)e^{-j\omega l}$$

- Most stationary random processes have continuous spectra
- If we have time, will discuss line spectra at end of term
- Recall the definition of PSD above
- The estimation problem is to find a $\hat{R}_x(e^{j\omega})$ given only a finite data record $\{x(n)\}_0^{N-1}$
- If the autocorrelation can be estimated with great accuracy at long lags, can treat as a deterministic problem
- With short records (or equivalently, local stationarity), is more difficult

Example 1: Data Windowed Autocorrelation

Solve for the expected value of the biased estimate of autocorrelation applied to a windowed data segment. How should $w(n)$ be scaled such that the estimate is unbiased at $\ell = 0$.

$$\hat{r}_x(\ell) = \frac{1}{N} \sum_{n=N-|\ell|-1}^{\infty} x(n+|\ell|)w(n+|\ell|)x^*(n)w^*(n)$$

Periodogram

$$\hat{R}_x(e^{j\omega}) \triangleq \frac{1}{N} \left| \sum_{n=0}^{N-1} v(n) e^{-j\omega n} \right|^2 = \frac{1}{N} |V(e^{j\omega})|^2$$

where $v(n) \triangleq x(n)w(n)$.

- If $w(n) = c$ (a constant), is called the **Periodogram**
- If $w(n)$ is not constant, is called the **Modified Periodogram** and $w(n)$ is called the **data window**
- Can be estimated quickly using the FFT
- Is related to the biased autocorrelation estimate we discussed last time

$$\hat{R}_x(e^{j\omega}) = \sum_{\ell=-(N-1)}^{N-1} \hat{r}_v(\ell) e^{-j\omega \ell}$$

Example 2: Periodogram and Biased Autocorrelation

Let $x(n)$ be a finite duration sequence, 0 to $N - 1$. Show that the biased estimate of autocorrelation is given by

$$\hat{r}_x(\ell) = \frac{1}{N} x(\ell) * x(-\ell)$$

and that the DTFT of $\hat{r}_x(\ell)$ is equal to the Periodogram of $x(n)$ when $w(n) = 1$.

Example 2: Workspace

Periodogram

$$\hat{R}_x(e^{j\omega}) \triangleq \frac{1}{N} \left| \sum_{n=0}^{N-1} v(n) e^{-j\omega n} \right|^2 = \frac{1}{N} |V(e^{j\omega})|^2 = \mathcal{F} \{ \hat{r}_v(\ell) \}$$

- Remember that we must use the biased estimate of autocorrelation
 - Otherwise the estimated PSD may be negative at some frequencies
 - The equality $\frac{1}{N} |V(e^{j\omega})|^2 = \mathcal{F} \{ \hat{r}_v(\ell) \}$ no longer holds
- This relationship to $\hat{r}(\ell)$ means we can use the FFT to calculate $\hat{r}(\ell)$ very efficiently
 - Take FFT of signal, magnitude square, and inverse FFT
 - Requires $\mathcal{O}(N \log N)$ operations instead of $\mathcal{O}(N^2)$!
 - Must take care to zero pad sufficiently

Example 3: Periodogram as a Filter Bank

$$\hat{R}_x(e^{j\omega}) \triangleq \frac{1}{N} \left| \sum_{n=0}^{N-1} v(n) e^{-j\omega n} \right|^2$$

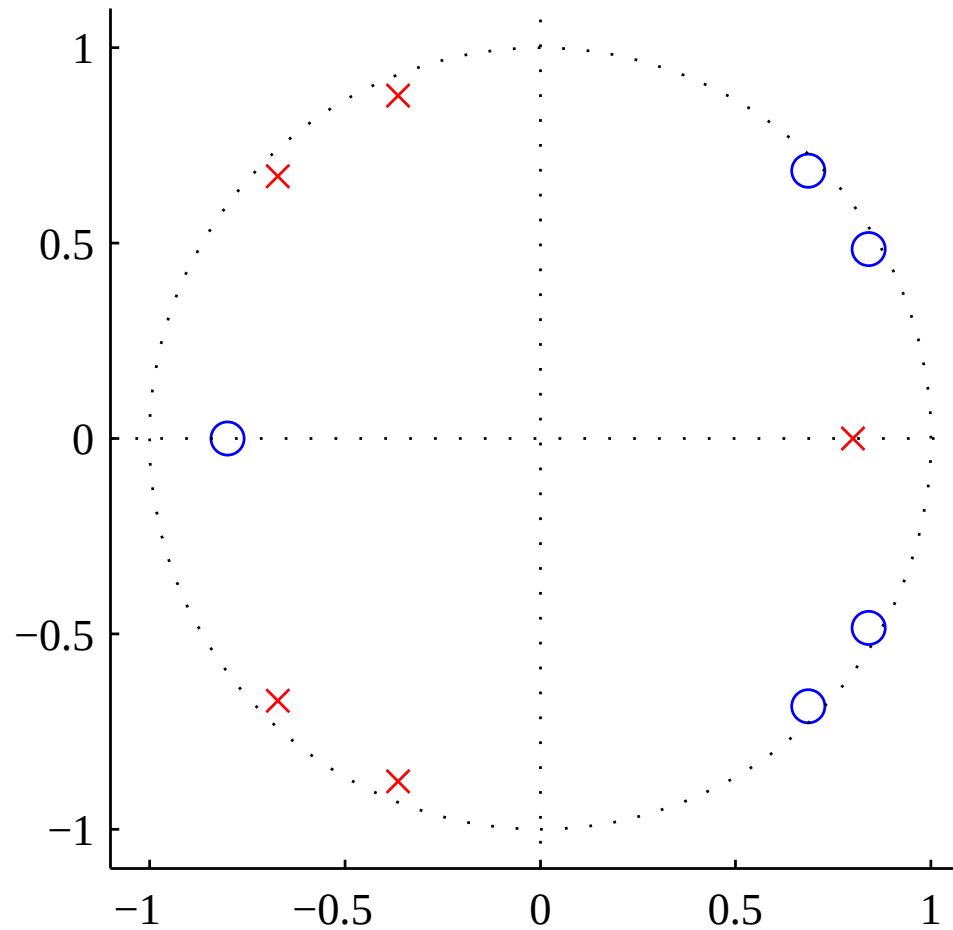
- How is the periodogram a filter bank?
- What is the transfer function of the filter?
- How good are the filters? How close are they to ideal filters?

Example 3: Work Space

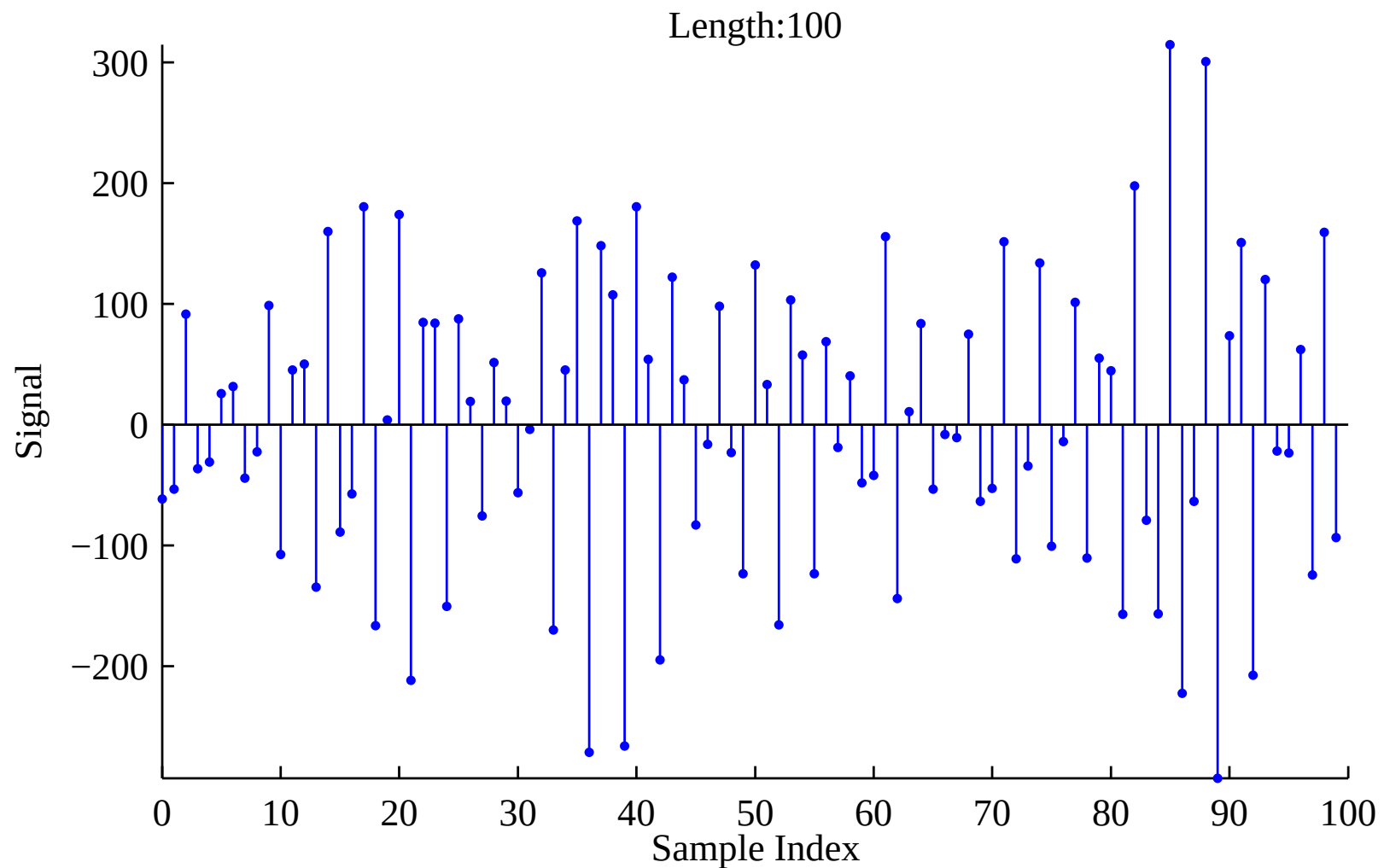
Example 4: Peridogram

Generate the periodogram for a signal from a known LTI system with two sets of complex conjugate poles close to one another and near the unit circle. Repeat for various signal lengths.

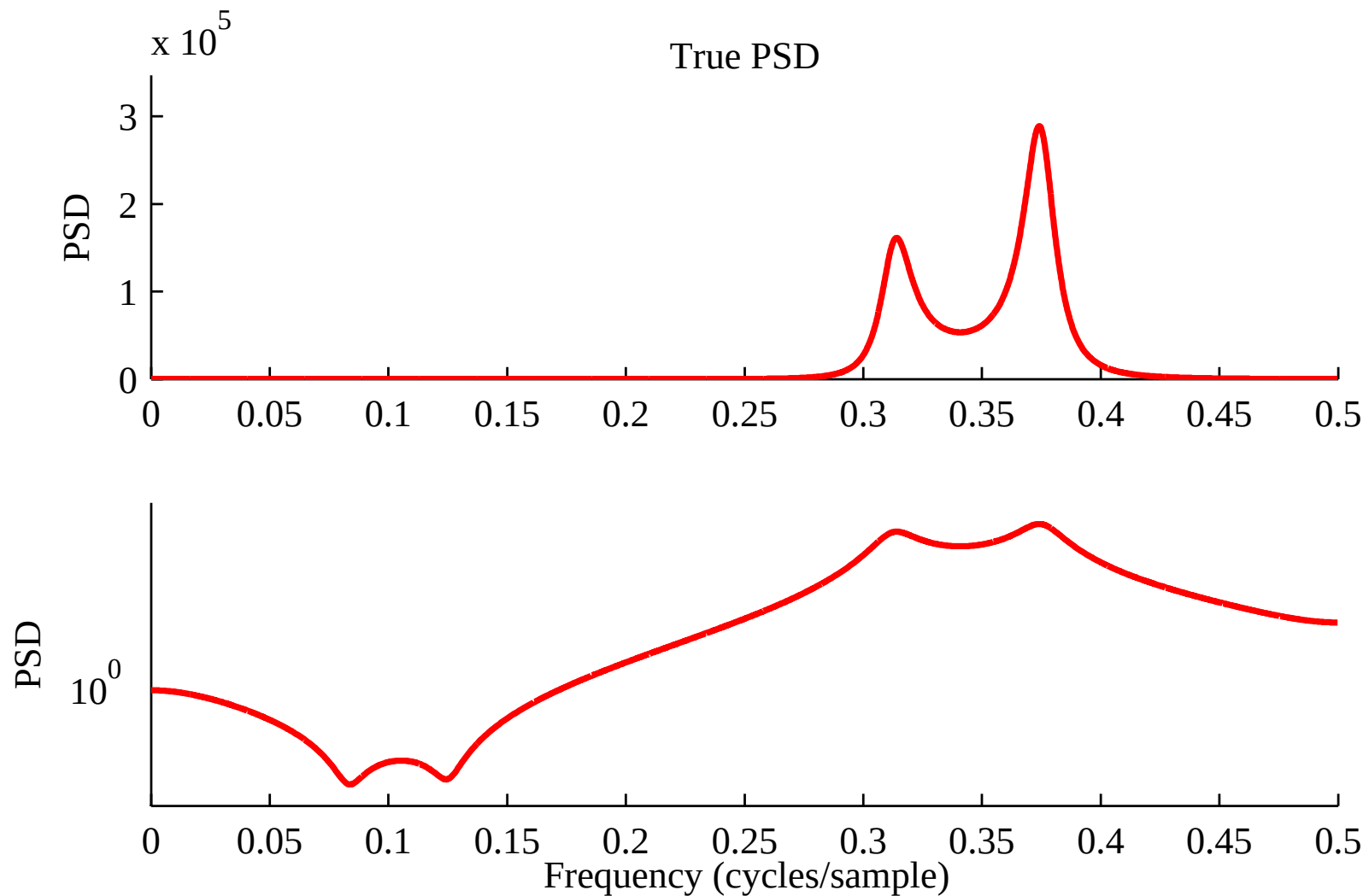
Example 4: Pole Zero Map of ARMA Process



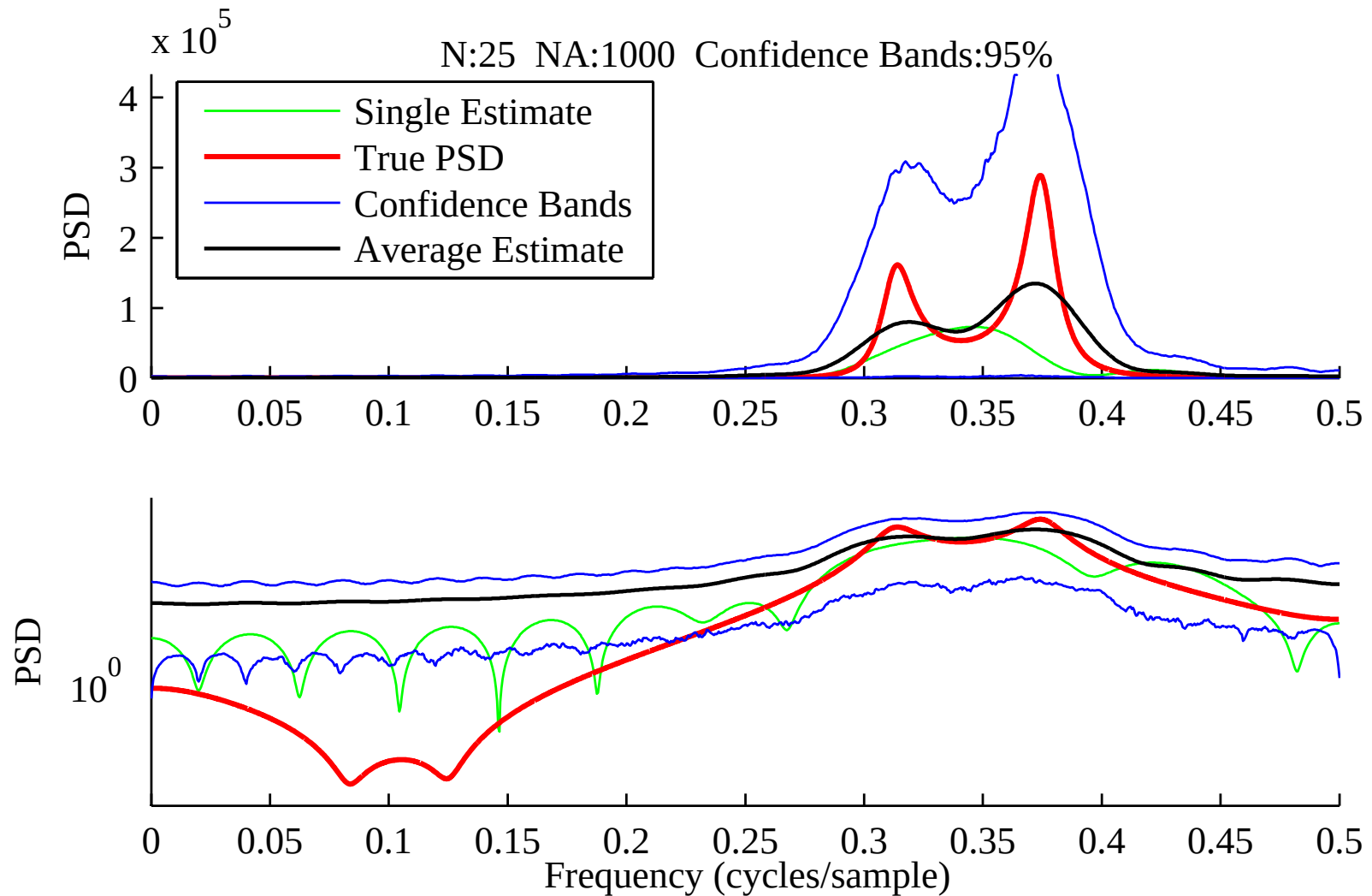
Example 4: Signal



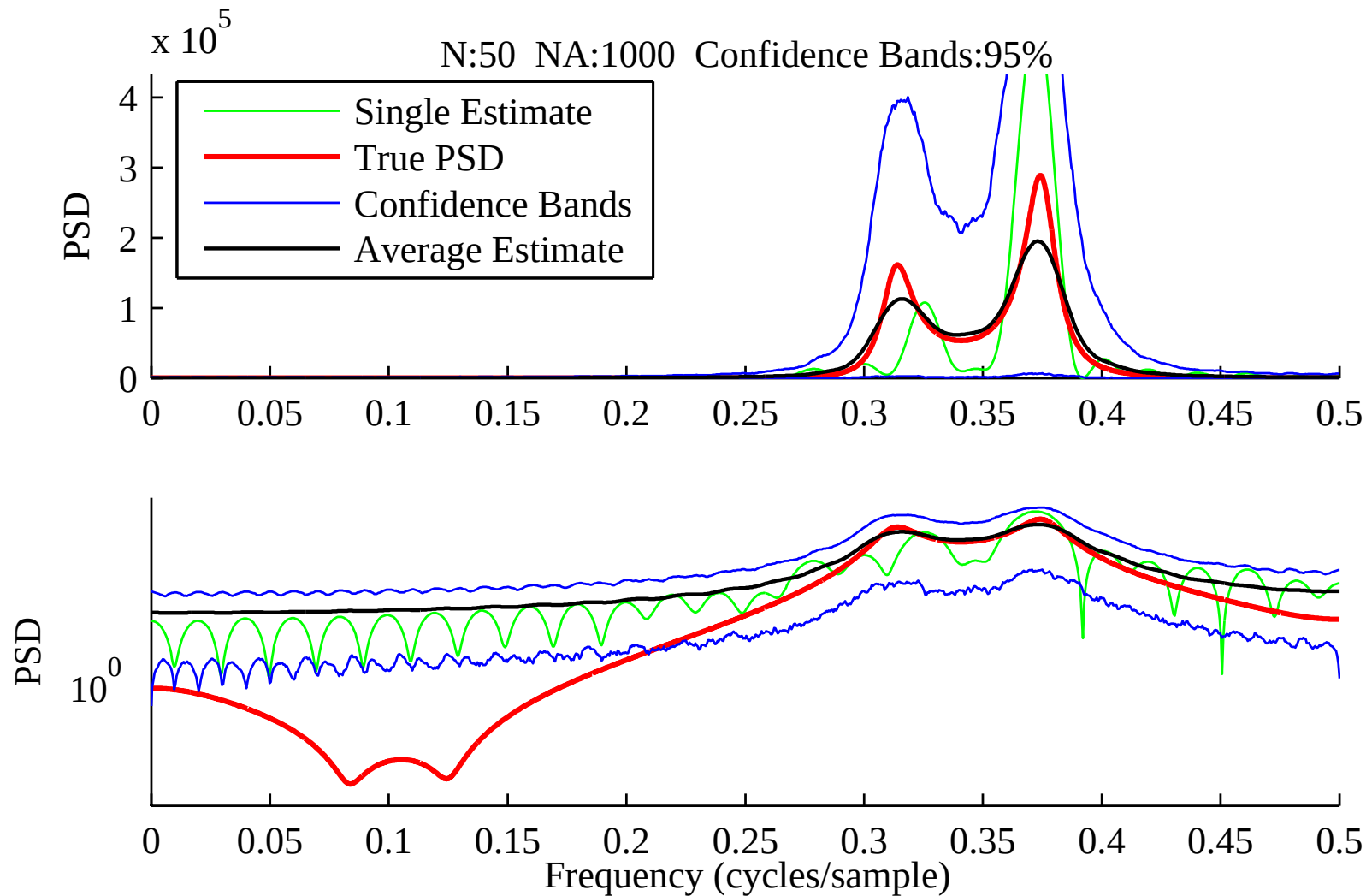
Example 4: True PSD



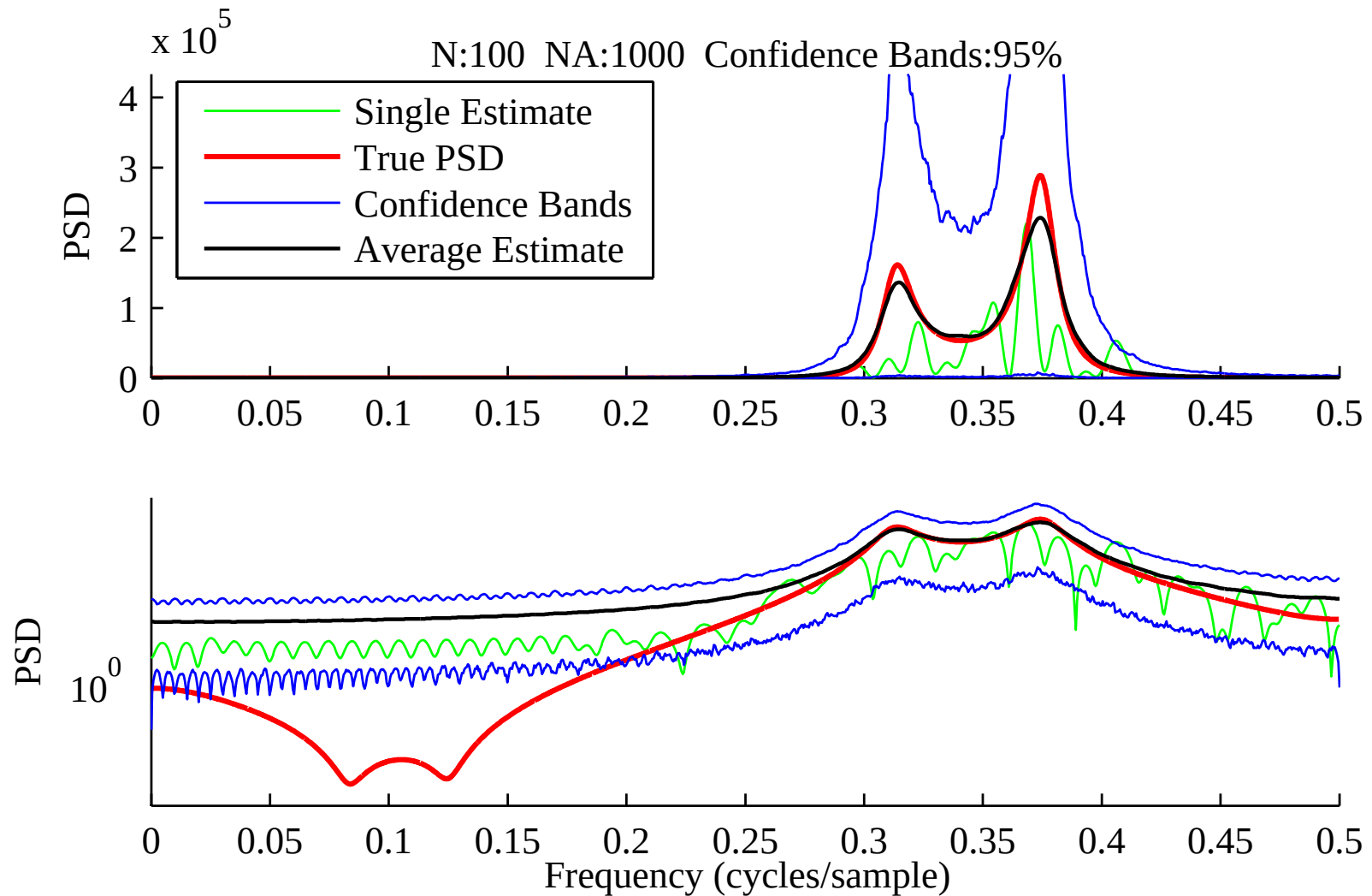
Example 4: Periodogram Estimate



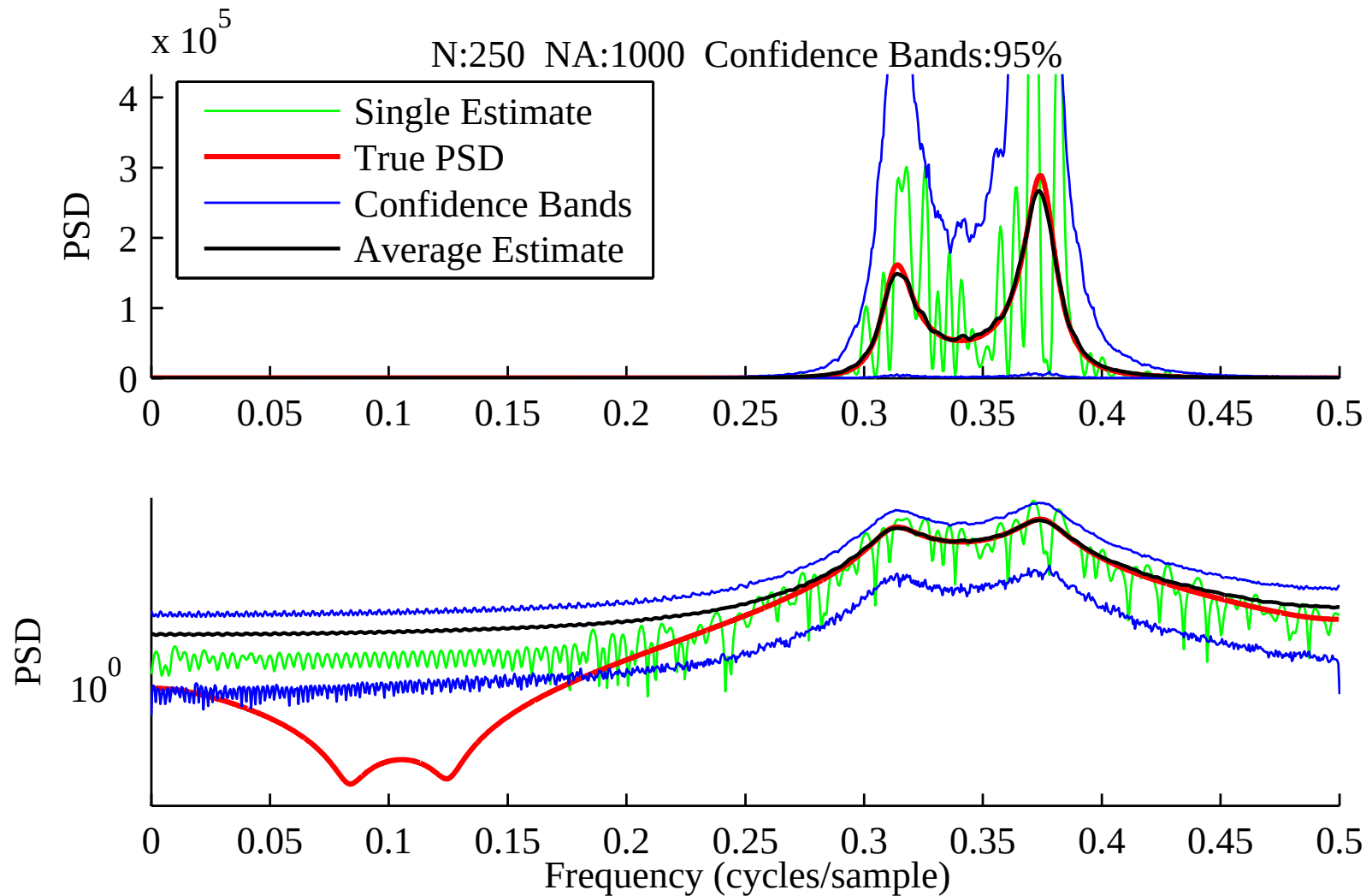
Example 4: Periodogram Estimate



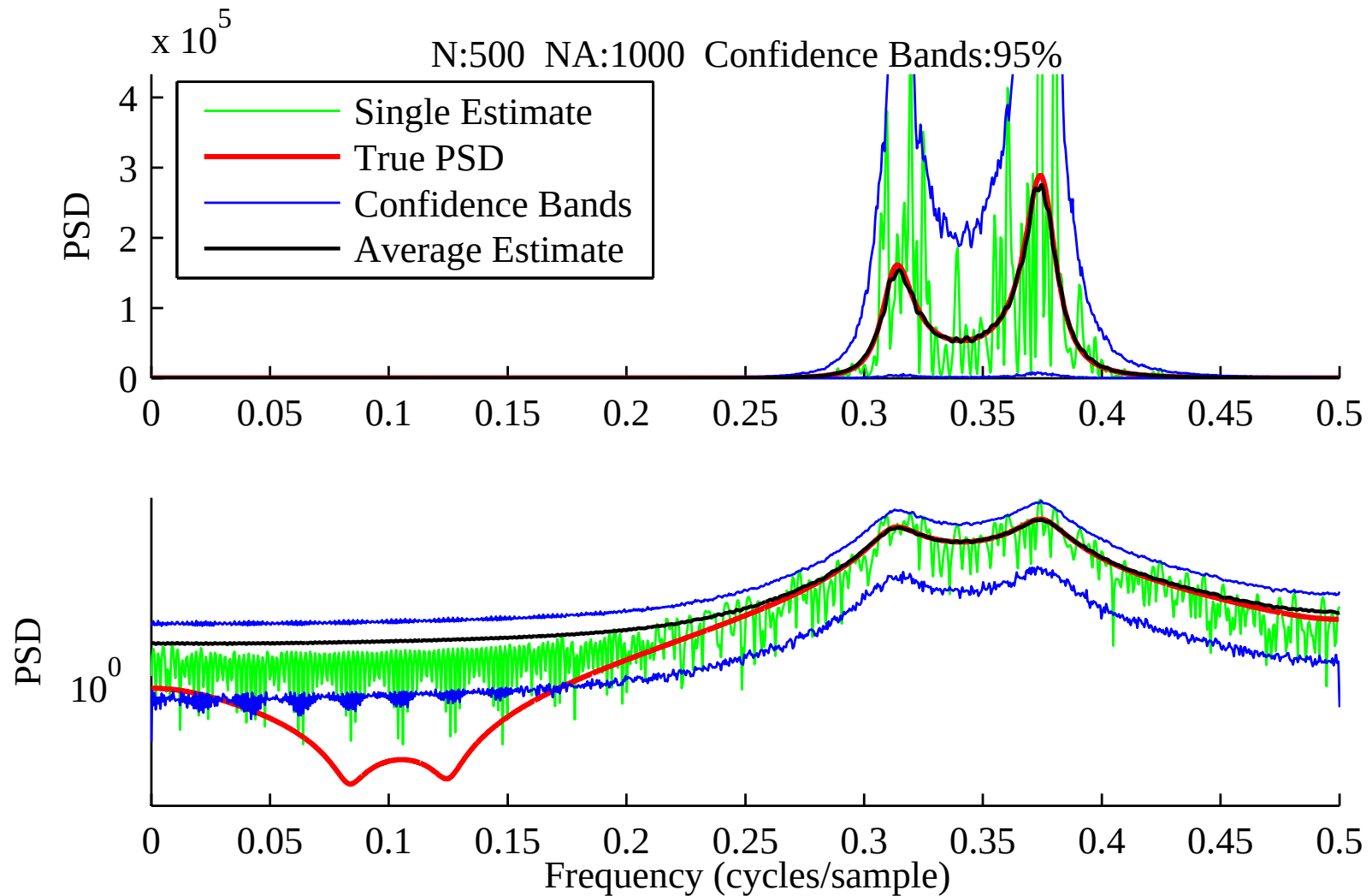
Example 4: Periodogram Estimate



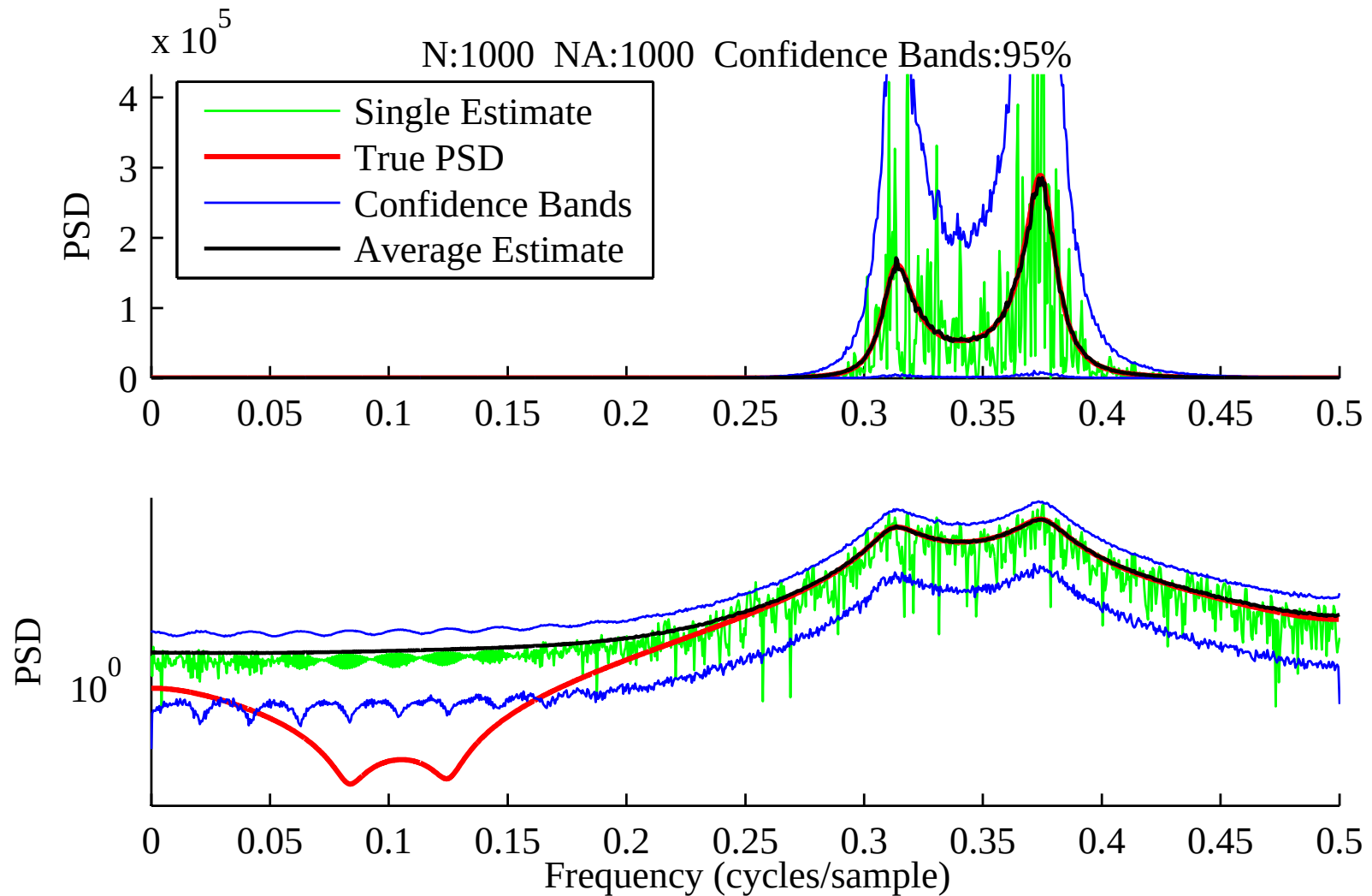
Example 4: Periodogram Estimate



Example 4: Periodogram Estimate



Example 4: Periodogram Estimate



Example 4: MATLAB Code

```
%function [] = Windows;
close all;
clear all;

M = 500; % Padding to eliminate transient
N = [25,50,100,250,500,1000]; % Length of signal segment
NA = 1000; % No. averages
cb = 95; % Confidence Bands
NZ = 2048;

b = poly([-0.8,0.97*exp(j*pi/4),0.97*exp(-j*pi/4),...
          0.97*exp(j*pi/6),0.97*exp(-j*pi/6)]); % Numerator coefficients
a = poly([ 0.8,0.95*exp(j*3.0*pi/4),0.95*exp(-j*3.0*pi/4),...
          0.95*exp(j*2.5*pi/4),0.95*exp(-j*2.5*pi/4)]); % Denominator coefficients
b = b*sum(a)/sum(b); % Scale DC gain to 1

%=====
% Plot the Pole-Zero Map
%=====
figure
FigureSet(1,'LTX');
h = Circle;
z = roots(b);
p = roots(a);
hold on;
h2 = plot(real(z),imag(z),'bo',real(p),imag(p),'rx');
hold off;
```

```

axis square;
xlim([-1.1 1.1]);
ylim([-1.1 1.1]);
AxisLines;
box off;
AxisSet(8);
print -depssc PeriodogramPZMap;

%=====
% Plot Signal
%=====

n = 100;
w = randn(M+n,1);
x = filter(b,a,w); % System with known PSD
nx = length(x);
x = x(nx-n+1:nx); % Eliminate start-up transient (make stationary)

figure;
FigureSet(1,'LTX');
k = 0:n-1;
h = stem(k,x);
set(h,'Marker','.');
set(h,'LineWidth',0.5);
box off;
ylabel('Signal');
xlabel('Sample Index');
title(sprintf('Length:%d',n));
xlim([0 n]);
ylim([min(x) max(x)]);
AxisSet(8);
print -depssc PeriodogramSignal;

```

```

%=====
% Plot True PSD
%=====

figure;
FigureSet(1,'LTX');
[R,w] = freqz(b,a,NZ);
R2 = abs(R).^2;
f = w/(2*pi);
subplot(2,1,1);
    h = plot(f,R2,'r');
    set(h,'LineWidth',1.3);
    box off;
    ylabel('PSD');
    title('True PSD');
    xlim([0 0.5]);
    ylim([0 max(R2)*1.2]);
subplot(2,1,2);
    h = semilogy(f,R2,'r');
    set(h,'LineWidth',1.3);
    box off;
    ylabel('PSD');
    xlim([0 0.5]);
    ylim([0.2*min(R2) max(R2)*5]);
xlabel('Frequency (cycles/sample)');
AxisSet(8);
print -depsc PeriodogramPSD;

%=====
% Plot Periodogram & Its Variance
%=====

```

```

for c1 = 1:length(N),
    nx = M+N(c1);
    Rh = zeros(NA,NZ/2+1);
    kh = 1:NZ/2+1;
    fh = (kh-1)/NZ;
    for c2 = 1:NA,
        w = randn(M+N(c1),1);
        x = filter(b,a,w); % System with known PSD
        x = x(nx-N(c1)+1:nx); % Eliminate start-up transient (make stationary)
        rh = (1/N(c1))*abs(fft(x,NZ)).^2;
        Rh(c2,:) = rh(kh).';
    end;

Rha = mean(Rh); % Average
Rhu = prctile(Rh,100-(100-cb)/2); % Upper confidence band
Rhl = prctile(Rh,(100-cb)/2); % Lower confidence band

figure;
FigureSet(1,'LTX');
subplot(2,1,1);
h = plot(fh,Rh(1,:), 'g',f,R2, 'r',fh,Rhl, 'b',fh,Rhu, 'b',fh,Rha, 'k');
set(h(1), 'LineWidth',0.5);
set(h(2), 'LineWidth',1.1);
set(h(3:4), 'LineWidth',0.5);
set(h(5), 'LineWidth',0.8);
box off;
ylabel('PSD');
title(sprintf('N:%d NA:%d Confidence Bands:%d%%',N(c1),NA,cb));
xlim([0 0.5]);
ylim([0 max(R2)*1.5]);
AxisSet(8);

```

```

        legend(h([1 2 4 5]),'Single Estimate','True PSD','Confidence Bands','Average Estimate',2);
subplot(2,1,2);
h = semilogy(fh,Rh(1,:), 'g',f,R2, 'r',fh,Rh1, 'b',fh,Rhu, 'b',fh,Rha, 'k');
set(h(1), 'LineWidth',0.5);
set(h(2), 'LineWidth',1.1);
set(h(3:4), 'LineWidth',0.5);
set(h(5), 'LineWidth',0.8);
box off;
ylabel('PSD');
xlabel('Frequency (cycles/sample)');
xlim([0 0.5]);
ylim([0.2*min(R2) max(R2)*5]);
AxisSet(8);
print(sprintf('PeriodogramAverage%04d',N(c1)),'-depsc');

Rhv = var(Rh); % Actual estimated ensemble variance
Rhvt = R2.^2; % Theoretical variance
figure;
FigureSet(2,'LTX');
subplot(2,1,1);
h = plot(fh,Rhv, 'r',f,Rhvt, 'g');
set(h(1), 'LineWidth',1.1);
set(h(2), 'LineWidth',0.5);
box off;
ylabel('var[PSD]');
title(sprintf('N:%d NA:%d Confidence Bands:%d%%',N(c1),NA,cb));
xlim([0 0.5]);
ylim([0 max(R2.^2)*1.25]);
AxisSet(8);
legend(h([1 2]),'Estimated Variance','Theoretical Variance',2);
subplot(2,1,2);

```

```

        h = semilogy(fh,Rhv,'r',f,Rhvt,'g');
        set(h(1) , 'LineWidth',1.1);
        set(h(2) , 'LineWidth',0.5);
        box off;
        ylabel('var[PSD]');
        xlabel('Frequency (cycles/sample)');
        xlim([0 0.5]);
        ylim([0.5*min(R2.^2) max(R2.^2)*2]);
AxisSet(8);
print(sprintf('PeriodogramVariance%04d',N(c1)),'-depsc');

end;

return;

for c1 = 1:length(N),
    w = randn(M+N(c1),1);
    x = filter(b,a,w); % System with known PSD
    nx = length(x);
    x = x(nx-M+1:nx); % Eliminate start-up transient (make stationary)

    Rh = (1/N(c1))*abs(fft(x,NZ)).^2;
    kh = 1:NZ/2+1;
    fh = (kh-1)/NZ;

    figure;
    FigureSet(1,'LTX');
    subplot(2,1,1);
        h = plot(fh,Rh(kh),'g',f,R2,'r');
        set(h(1),'LineWidth',0.5);
        set(h(2),'LineWidth',1.2);

```

```

        box off;
        ylabel('PSD');
        title(sprintf('N:%d',N(c1)));
        xlim([0 0.5]);
        ylim([0 max(R2)*1.2]);
subplot(2,1,2);
    h = semilogy(fh,Rh(kh),'g',f,R2,'r');
    set(h(1),'LineWidth',0.5);
    set(h(2),'LineWidth',1.2);
    box off;
    ylabel('PSD');
    xlabel('Frequency (cycles/sample)');
    xlim([0 0.5]);
    ylim([0.2*min(R2) max(R2)*5]);
AxisSet(8);
print(sprintf('PeriodogramEstimate%04d',N(c1)),'-depsc');
end;

```

Periodogram Properties

- The main disadvantage of the Periodogram appears to be excessive variance
- As with any estimator, we would like to know the bias, variance, covariance, confidence intervals, and distribution
- We will consider these each in turn

Periodogram Mean

$$\begin{aligned} \mathbb{E} \left[\hat{R}_x(e^{j\omega}) \right] &= \sum_{\ell=-(N-1)}^{N-1} \mathbb{E} [\hat{r}_v(\ell)] e^{-j\omega\ell} \\ \mathbb{E} [\hat{r}_v(\ell)] &= \mathbb{E} \left[\frac{1}{N} \sum_{n=-\infty}^{\infty} x(n)w(n)x^*(n-\ell)w^*(n-\ell) \right] \\ &= \frac{1}{N} \sum_{n=-\infty}^{\infty} r_x(\ell)w(n)w^*(n-\ell) \\ &= r_x(\ell) \frac{1}{N} \sum_{n=-\infty}^{\infty} w(n)w^*(n-\ell) \\ &= \frac{1}{N} r_x(\ell)r_w(\ell) \end{aligned}$$

The expected value of $\hat{r}_v(\ell)$ is the true autocorrelation $r_x(\ell)$ windowed with $r_w(\ell) = \frac{1}{N} w(n) * w(-n)^*$.

Periodogram Mean Continued

$$\begin{aligned} \mathbb{E} \left[\hat{R}_x(e^{j\omega}) \right] &= \sum_{\ell=-(N-1)}^{N-1} \mathbb{E} [\hat{r}_x(\ell)] e^{-j\omega\ell} \\ &= \frac{1}{N} \sum_{\ell=-(N-1)}^{N-1} r_x(\ell) r_w(\ell) e^{-j\omega\ell} \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} R_x(e^{ju}) \frac{1}{N} R_w(e^{j(\omega-u)}) du \end{aligned}$$

- Since $\mathbb{E} \left[\hat{R}_x(e^{j\omega}) \right] \neq R_x(e^{j\omega})$, the Periodogram is a biased estimator
- Here $R_w(e^{j\omega}) = |W(e^{j\omega})|^2$ is the ESD of the window
- Ideally, we would like $R_w(e^{j\omega}) = 2\pi N \delta(\omega)$
- The smearing limits our ability to distinguish sharp features (e.g. distinct peaks) in the PSD

Periodogram Asymptotic Bias

$$\mathbb{E} \left[\hat{R}_x(e^{j\omega}) \right] = \frac{1}{N} \sum_{\ell=-(N-1)}^{N-1} r_x(\ell) r_w(\ell) e^{-j\omega\ell}$$

- Most windows are relatively constant over a range of samples proportional to the window length:

$$r_w(\ell) \approx \begin{cases} N & |\ell| \ll N \\ 0 & |\ell| > N \end{cases}$$

- Most stationary signals have an autocorrelation that approaches zero as $\ell \rightarrow \infty$
- Thus, the periodogram can be considered to be an asymptotically unbiased estimator

$$\lim_{N \rightarrow \infty} \mathbb{E} \left[\hat{R}_x(e^{j\omega}) \right] = R_x(e^{j\omega})$$

Periodogram Asymptotic Bias Continued

$$\lim_{N \rightarrow \infty} \mathbb{E} \left[\hat{R}_x(e^{j\omega}) \right] = R_x(e^{j\omega})$$

- The estimator is unbiased as long as

$$\sum_{n=0}^{N-1} |w(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} R_w(e^{j\omega}) d\omega = N$$

and the mainlobe window width decreases as $\frac{1}{N}$

- Not of much relevance (asymptotic result)
- Practically, the bias is introduced by the sidelobes and smearing of the spectrum by the main lobe
- Can reduce with better windows, but can only trade main lobe width for decreased sidebands
- If signal is white noise (flat spectrum), the smearing of the convolution does not cause bias

Periodogram Covariance Theory

$$\hat{R}_w(e^{j\omega}) = \frac{1}{N} \left| \sum_{n=0}^{N-1} w(n) e^{j\omega n} \right|^2 = \frac{1}{N} \left| \sum_{n=0}^{N-1} w(n) e^{jk \frac{2\pi}{N} n} \right|^2$$

- The text lists the covariance of the Periodogram, but does not derive it
- Details of the derivation can be found in [?]
- An outline of the derivation is provided here
- Suppose for now that no zero padding is used so we calculate the DFT only at the N points that we have samples of the signal
- Let $w(n)$ be a WGN process with variance σ_w^2 and zero mean

Orthogonal Components

Recall that when zero padding is not used, the exponentials are orthogonal

$$\begin{aligned}\sum_{n=\langle N \rangle} \left(e^{jk \frac{2\pi}{N} n} \right) \left(e^{j\ell \frac{2\pi}{N} n} \right)^* &= \sum_{n=\langle N \rangle} e^{j(k-\ell) \frac{2\pi}{N} n} \\ &= \begin{cases} N & k = 0, \pm N, \pm 2N, \dots \\ 0 & \text{otherwise} \end{cases}\end{aligned}$$

Now consider $W(e^{j\omega})$ evaluated at $\omega = k \frac{2\pi}{N}$.

$$W(e^{j\omega}) = \sum_{n=0}^{N-1} w(n) e^{j\omega n}$$

If $w(n)$ is a WGN process, then $W(e^{j\omega})$ is also a Gaussian random variable, since it is a linear combination of Gaussian RVs.

WGN PSD Mean and Variance

$$\text{E} [W(e^{j\omega})] = \sum_{n=0}^{N-1} \text{E}[w(n)]e^{j\omega n} = 0$$

$$\begin{aligned} \text{cov} [W(e^{j\omega_1}), W(e^{j\omega_2})] &= \text{E} [W(e^{j\omega_1})W(e^{j\omega_2})^*] \\ &= \text{E} \left[\left(\sum_{k=0}^{N-1} w(k)e^{j\omega_1 k} \right) \left(\sum_{\ell=0}^{N-1} w(\ell)e^{j\omega_2 \ell} \right)^* \right] \\ &= \sum_{k=0}^{N-1} \sum_{\ell=0}^{N-1} \text{E}[w(k)w(\ell)^*]e^{j\omega_1 k}e^{-j\omega_2 \ell} \\ &= \sum_{k=0}^{N-1} \sum_{\ell=0}^{N-1} \sigma_w^2 \delta(k - \ell)e^{j(\omega_1 k - \omega_2 \ell)} \\ &= \sigma_w^2 \sum_{k=0}^{N-1} e^{j(\omega_1 - \omega_2)k} \end{aligned}$$

WGN DFT Mean and Variance Continued

If we don't use zero padding,

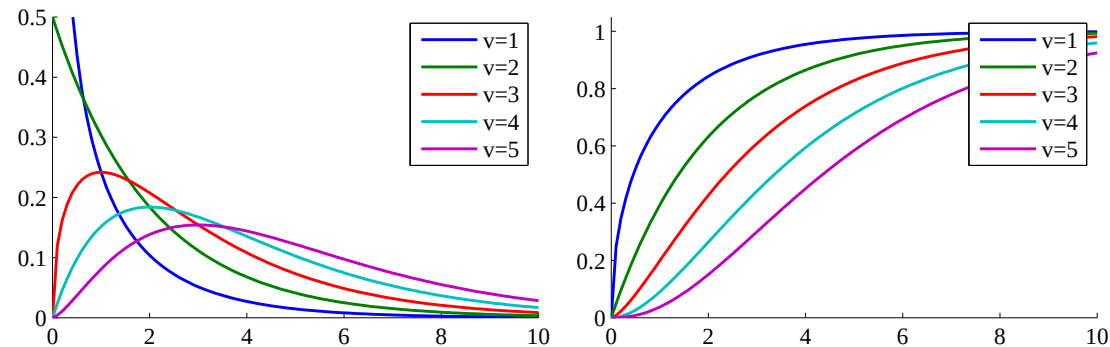
$$\omega_1 = m_1 \frac{2\pi}{N} \qquad \omega_2 = m_2 \frac{2\pi}{N}$$

where m_1 and m_2 are integers. Then

$$\begin{aligned} \text{cov} [W(e^{j\omega_1}), W(e^{j\omega_2})] &= \sigma_w^2 \sum_{k=0}^{N-1} e^{j(\omega_1 - \omega_2)k} \\ &= \sigma_w^2 \sum_{k=0}^{N-1} e^{j(m_1 - m_2) \frac{2\pi}{N} k} \\ &= \sigma_w^2 N \delta(m_1 - m_2) \end{aligned}$$

Thus, for WGN and no zero padding, the random variables are uncorrelated and therefore independent! The variance of the RVs is σ_w^2 . Alternative, the number of uncorrelated estimates in the frequency domain scales with N . More data $\rightarrow$ more uncorrelated samples.

χ^2 Random Variables



- Let x_i be a set of IID RVs with a Normal distribution, $x_i \sim \mathcal{N}(0, 1)$
- Then the RV χ^2 where

$$\chi^2 = \sum_{i=1}^v x_i^2$$

has a χ^2 distribution with v degrees of freedom

- The PDF and CDF is available in MATLAB and equivalent

MATLAB Code

```
clear;
close all;

%=====
% Plot the Pole-Zero Map
%=====

x = linspace(0,10,100)';
df = 1:5; % Degrees of freedom
nd = length(df);
nx = length(x);
P = zeros(nx,nd);
C = zeros(nx,nd);
for c1=1:5,
    P(:,c1) = chi2pdf(x,df(c1));
    C(:,c1) = chi2cdf(x,df(c1));
    ls{c1} = sprintf('v=%d',df(c1));
end;

figure
FigureSet(1,'LTX');
h = plot(x,P);
set(h,'LineWidth',1.5);
box off;
xlim([0 x(end)]);
ylim([0 .5]);
AxisSet(12);
```

```
legend(ls);  
print -depsc Chi2PDF;  
  
figure  
FigureSet(2,'LTX');  
h = plot(x,C);  
set(h,'LineWidth',1.5);  
box off;  
xlim([0 x(end)]);  
ylim([0 1.05]);  
AxisSet(12);  
legend(ls);  
print -depsc Chi2CDF;
```

WGN PSD Mean and Variance Continued

$$\mathbb{E} [W(e^{j\omega})] = \sum_{n=0}^{N-1} \mathbb{E}[w(n)]e^{j\omega n} = 0$$

$$\text{cov} [W(e^{j\omega_1}), W(e^{j\omega_2})] = \sigma_w^2 N \delta(\omega_1 - \omega_2)$$

$$\hat{R}_w(e^{j\omega}) = \frac{1}{N} |W(e^{j\omega})|^2$$

- If $W(e^{j\omega})$ is a complex Gaussian RV with zero mean and variance σ_w^2 , then what is the distribution of $|W(e^{j\omega})|^2$?

$$\mathbb{E} [|W(e^{j\omega})|^2] = \mathbb{E} [\text{Re}\{W(e^{j\omega})\}^2 + \text{Im}\{W(e^{j\omega})\}^2] = \sigma_w^2$$

where the real and imaginary components are both independent Gaussian RVs each with variance $\sigma_w^2/2$

- Note that $\text{Im}\{W(e^{j\omega})\} = 0$ if $\omega = 0$ or $\omega = \pi$

WGN PSD Distribution

Thus the distribution of $|W(e^{j\omega})|^2$ is given by

$$\frac{|W(e^{j\omega})|^2}{\sigma_w^2 N} \sim \begin{cases} \chi^2(2)/2 & \omega \neq \ell\pi \\ \chi^2(1) & \omega = \ell\pi \end{cases}$$

where $\chi^2(1)$ is a chi-squared distribution with one degree of freedom.
Thus

$$\hat{R}_w(e^{j\omega}) \sim \begin{cases} \sigma_w \chi^2(2)/2 & \omega \neq \ell\pi \\ \sigma_w \chi^2(1) & \omega = \ell\pi \end{cases}$$

Thus, we know what the distribution of $\hat{R}_w(e^{j\omega})$ is in the WGN case, but what about non-white processes?

Generalized Linear Processes

$$x(n) = h(n) * w(n) = \sum_{k=-\infty}^{\infty} h(k)w(n-k)$$

$$R_x(e^{j\omega}) = |H(e^{j\omega})|^2 \sigma_w^2$$

- This is a standard and widely used model for wide-sense stationary processes
- Note that $h(n)$ is two sided (non-causal)
- We require that $h(n)$ has finite energy (a sufficient condition for stability)
- But how is $\hat{R}_x(e^{j\omega})$ related to the known distribution of $\hat{R}_w(e^{j\omega})$?

Periodogram Distribution

It's beyond the scope of this class to prove it (see [?, pg. 424]), but

$$\hat{R}_x(e^{j\omega}) = R_x(e^{j\omega}) \frac{1}{\sigma_w^2} \hat{R}_w(e^{j\omega}) + R_\varepsilon(\omega)$$

where

$$E[R_\varepsilon(\omega)] = \mathcal{O}(1/N^2)$$

- Thus $R_\varepsilon(\omega) \rightarrow 0$ as $N \rightarrow \infty$
- If we assume N is “large” we can disregard it
 - N must be large enough to make the bias due to windowing negligible
- This means that the periodogram estimate is χ^2 distributed when the process is Gaussian
- We can then immediately write the distribution of $\hat{R}_x(e^{j\omega})$

Periodogram Distribution

$$\hat{R}_x(e^{j\omega}) \sim \begin{cases} R_x(e^{j\omega})\chi^2(2)/2 & \omega \neq \ell\pi \\ R_x(e^{j\omega})\chi^2(1) & \omega = \ell\pi \end{cases}$$

Since

$$\text{var} [\chi^2(2)/2] = 1 \qquad \text{var} [\chi^2(1)] = 2$$

The variance of the periodogram is given by

$$\text{var}\{\hat{R}_x(e^{j\omega})\} \approx \begin{cases} R_x^2(e^{j\omega}) & 0 < \omega < \pi \\ 2R_x^2(e^{j\omega}) & \omega = 0, \pi \end{cases}$$

which agrees with the text.

Periodogram Covariance

- Depends on the true, unknown PSD so is of limited value
- Can make some qualitative observations
- Estimates at two frequencies $\omega_1 = (2\pi/N)k_1$ and $\omega_2 = (2\pi/N)k_2$ where k_1 and k_2 are integers are approximately uncorrelated

$$\text{cov}\{\hat{R}_x(e^{j\omega_1}), \hat{R}_x(e^{j\omega_2})\} \approx 0 \text{ for } k_1 \neq k_2$$

- The number of uncorrelated frequency pairs scales with N
- Thus, the periodogram at adjacent frequencies, say ω and $\omega + \delta$, *becomes more variable* as N increases

Periodogram Variance (Again)

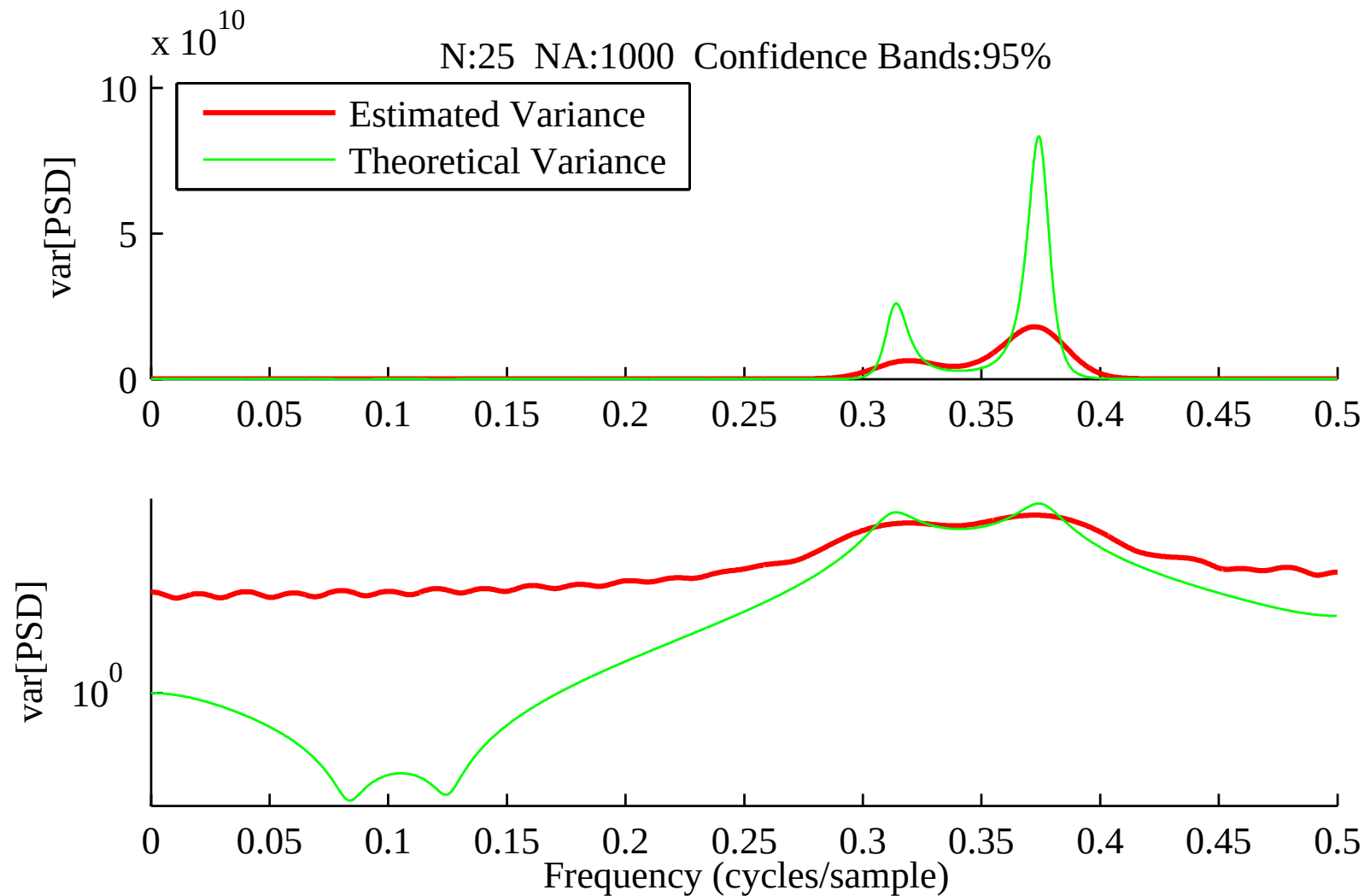
$$\text{var}\{\hat{R}_x(e^{j\omega})\} \approx R_x^2(e^{j\omega}) \left[1 + \left(\frac{\sin(\omega N)}{N \sin(\omega)} \right)^2 \right]$$

For large N this can be approximated as

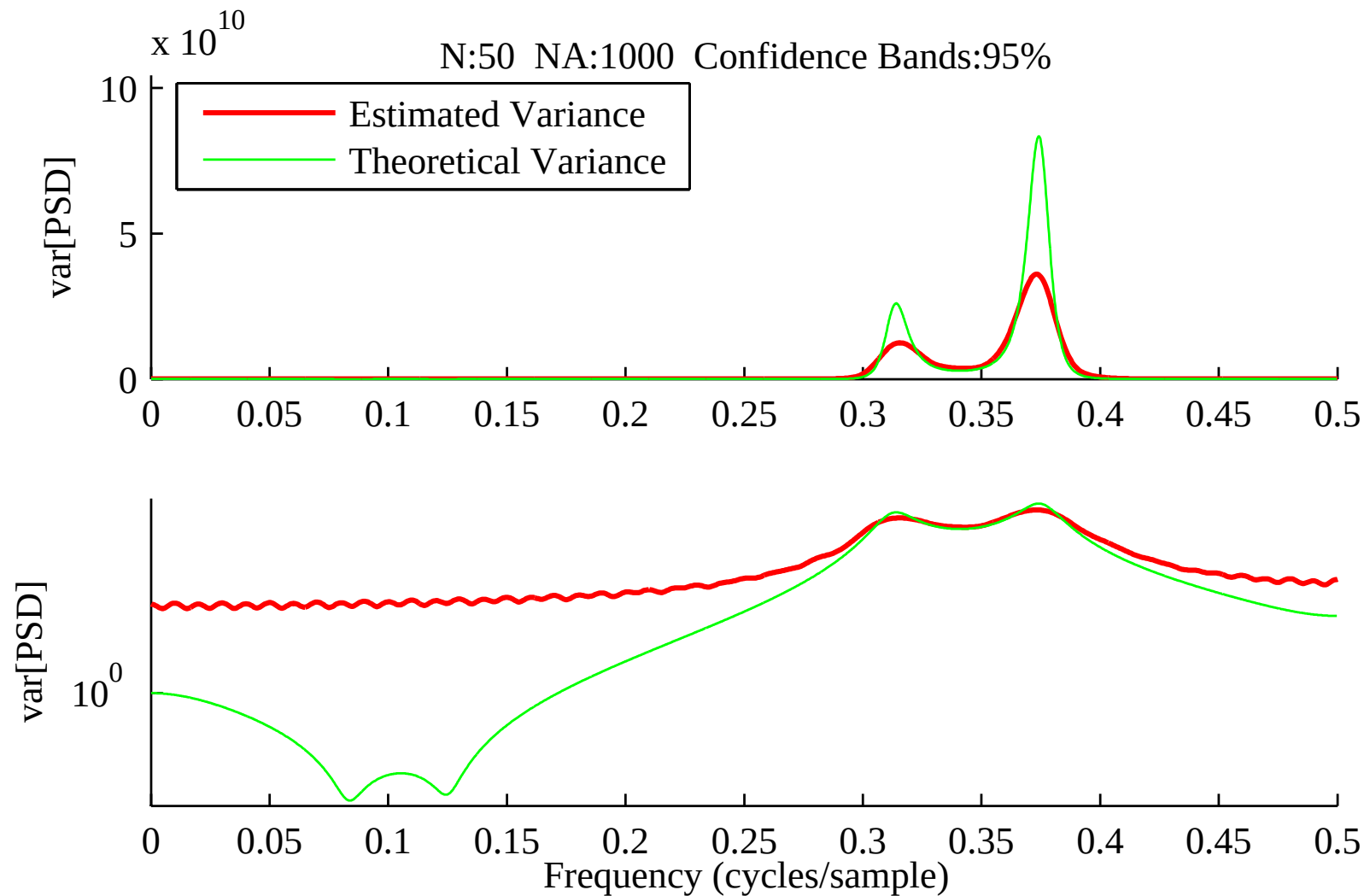
$$\text{var} \left[\hat{R}_x(e^{j\omega}) \right] \approx \begin{cases} R_x^2(e^{j\omega}) & 0 < \omega < \pi \\ 2R_x^2(e^{j\omega}) & \omega = 0, \pi \end{cases}$$

- The variance depends on the true spectrum
- Perhaps not surprising since the variance of $\hat{r}(\ell)$ depended on the true autocorrelation
- Note that the variance does not decrease as $N \rightarrow \infty$
- Thus, the Periodogram is not a consistent estimator
- The estimate becomes more variable as a function of frequency, but the variance at a fixed frequency does not decrease

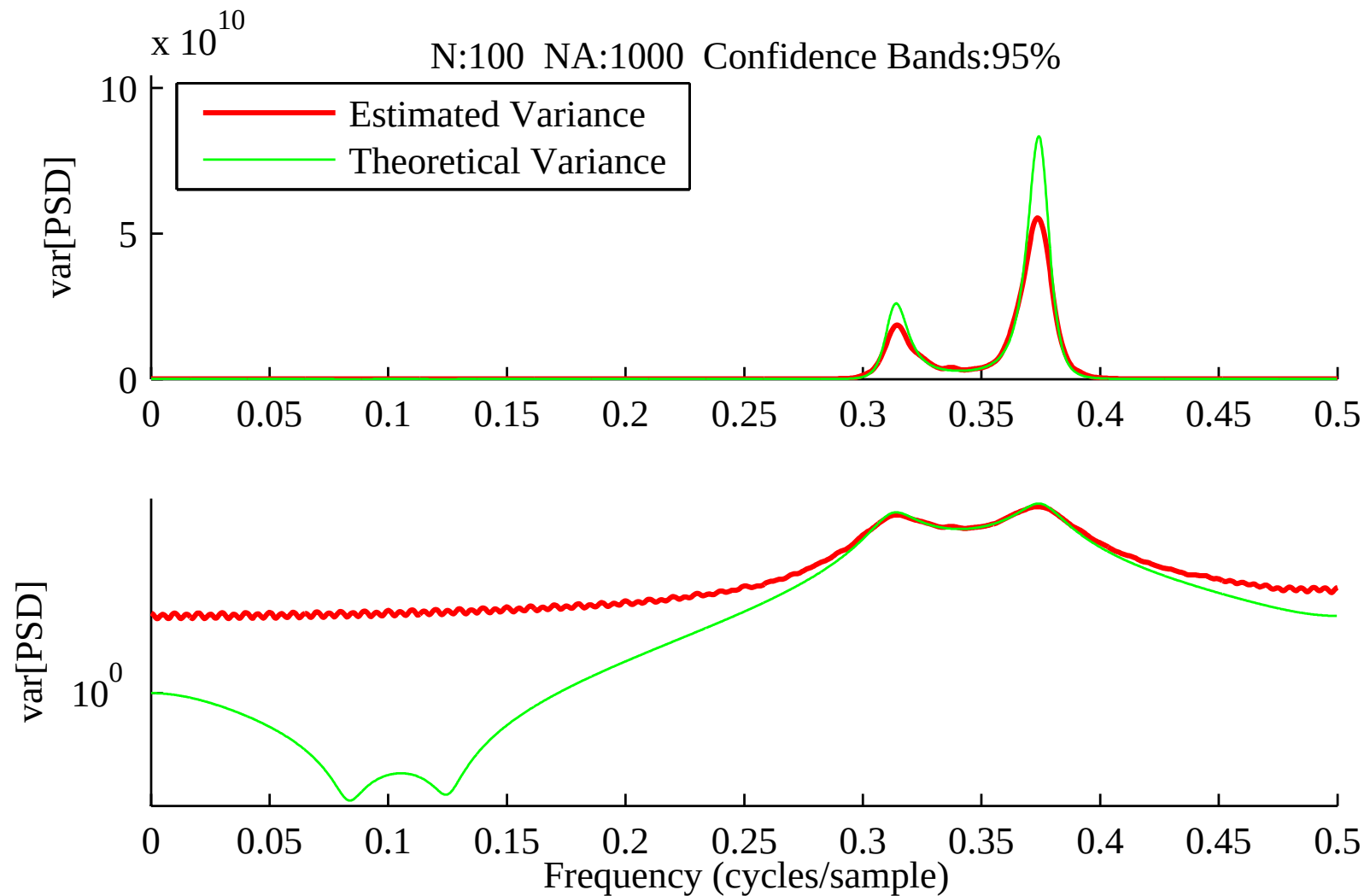
Example 4: Periodogram Variance



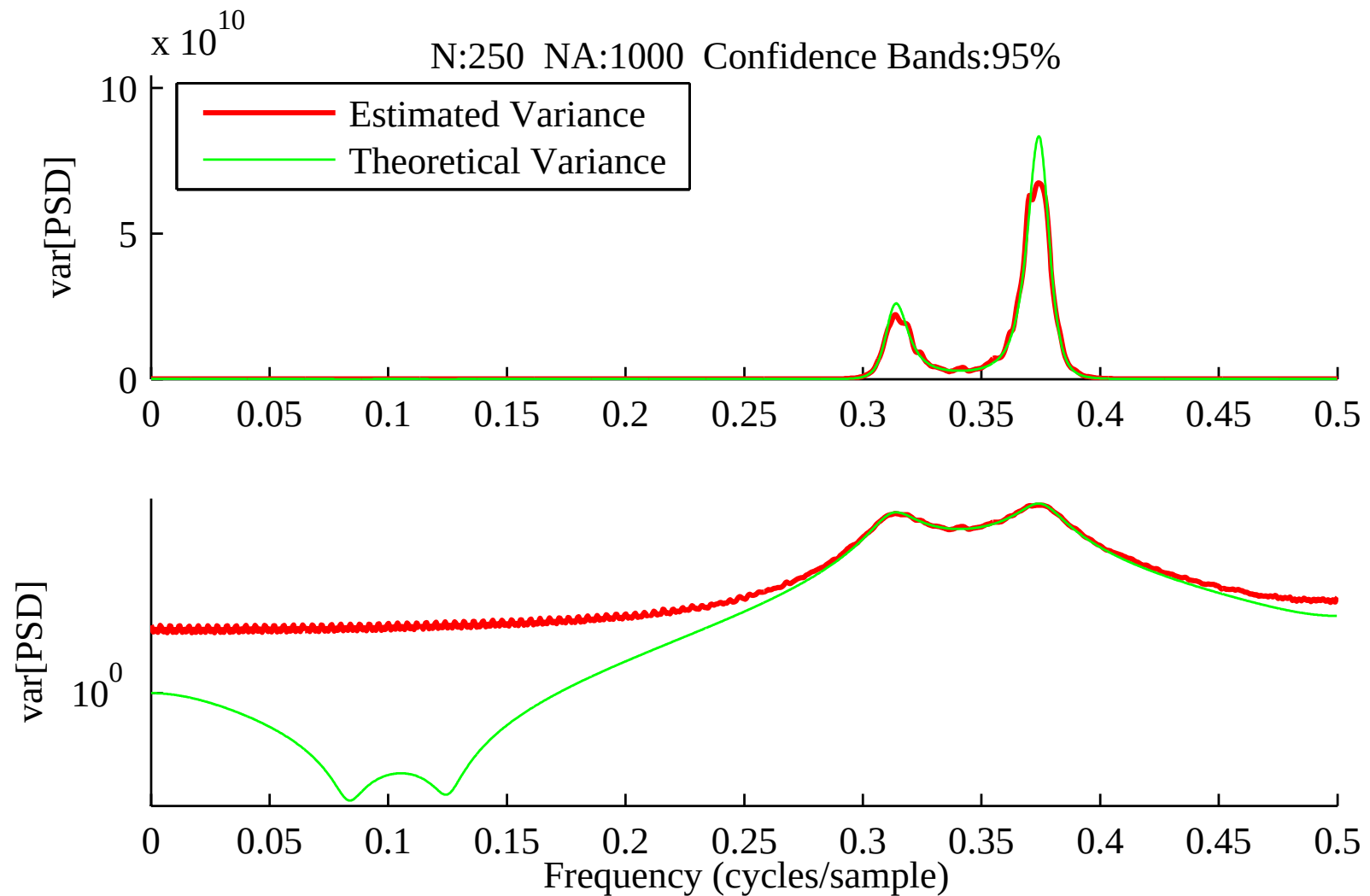
Example 4: Periodogram Variance



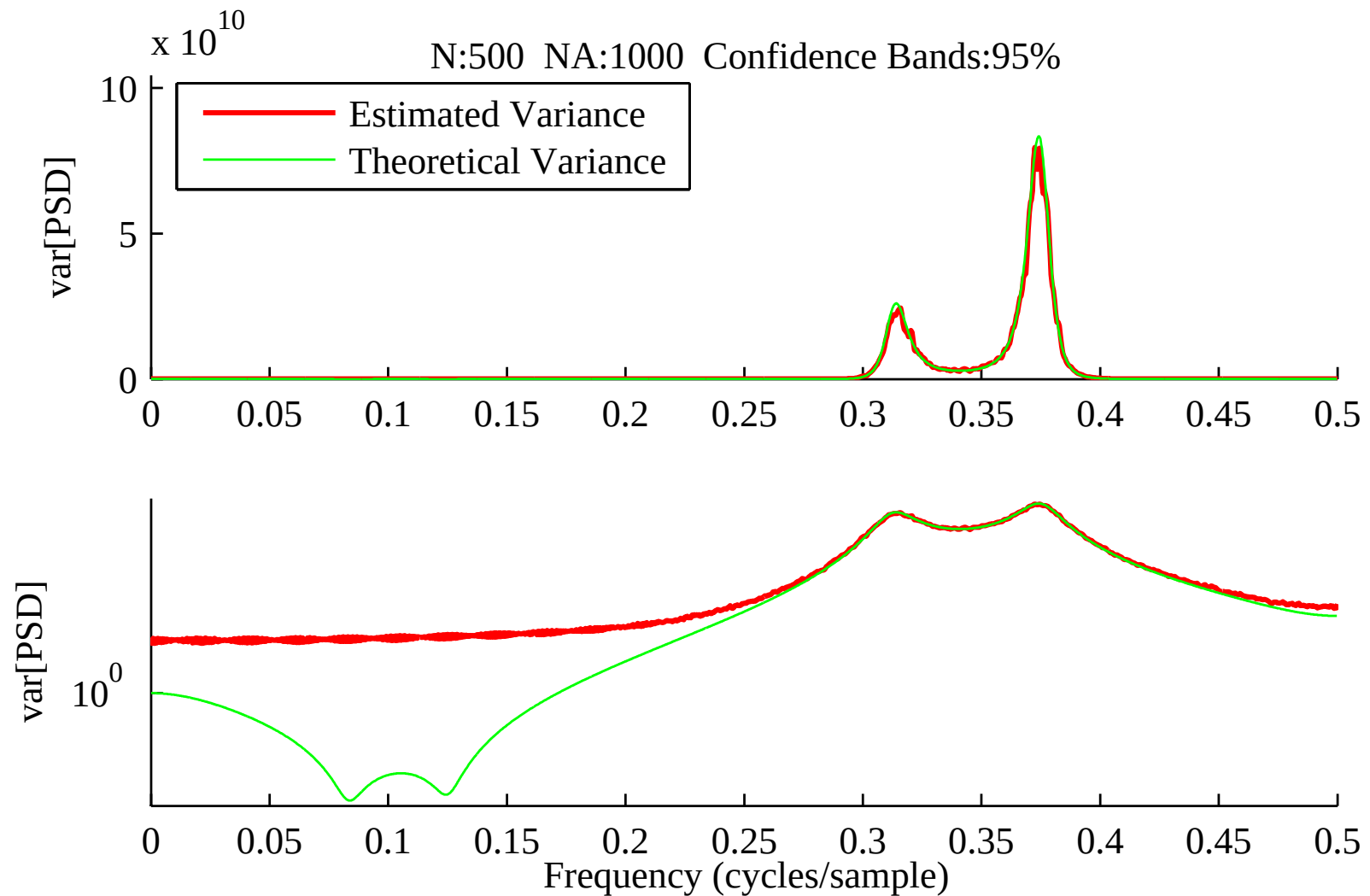
Example 4: Periodogram Variance



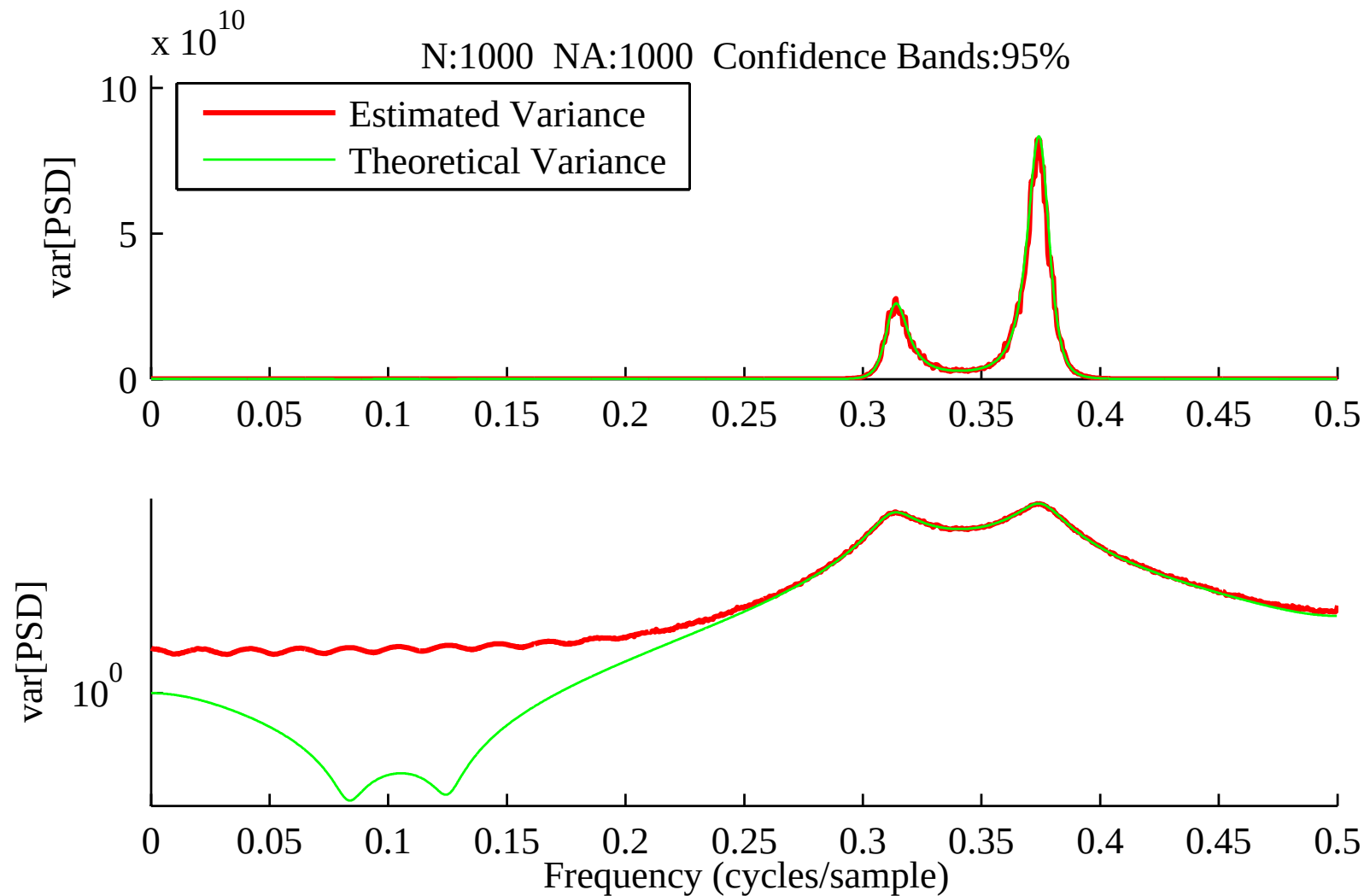
Example 4: Periodogram Variance



Example 4: Periodogram Variance



Example 4: Periodogram Variance



Periodogram Variance

$$\hat{R}_x(e^{j\omega}) \triangleq \frac{1}{N} \left| \sum_{n=0}^{N-1} v(n)e^{-j\omega n} \right|^2 = \frac{1}{N} |V(e^{j\omega})|^2 = \mathcal{F} \{ \hat{r}(\ell) \}$$

- Since the ensemble variance of the Periodogram does not decrease as N increases, it is considered a poor estimator
- We cannot fix this by choosing a better window
- This is perhaps surprising, because it is the “natural” estimator
- To obtain a better estimator, we need to reduce the ensemble variance

Fixing Problems with the Periodogram

- As N increases, the *ensemble* variation stays constant
 - The estimator is not consistent (statistically)
- As N increases, the covariation at neighboring frequencies, ω and $\omega + \delta$, decreases
- Thus we are essentially able to estimate the true PSD at more frequencies
- We have better frequency resolution as N increases
- Key assumption: the true PSD is smooth
 - Clearly true for all ARMA processes with poles and zeros not too close to the unit circle
- Key concept: If the PSD is smooth and our estimate consists of uncorrelated high-resolution estimates, we can obtain a better estimate by averaging estimates at adjacent frequencies

Periodogram Smoothing

- Smoothing the Periodogram can be thought of in several ways
 - It reduces the frequency resolution
 - It blurs the spectrum
 - It increases the bias of the estimator
 - It reduces the variance of the estimator
 - It could either increase or decrease the MSE depending on the degree of smoothing and the variability of the true PSD
- Thus, we can tradeoff some of the excessive variance of the Periodogram for increases bias by smoothing
- There are several ways to achieve this
 - Average contiguous values of the periodogram
 - Average periodograms obtained from multiple segments
- Both approaches produce about the same quality of estimators

Smoothing the Periodogram

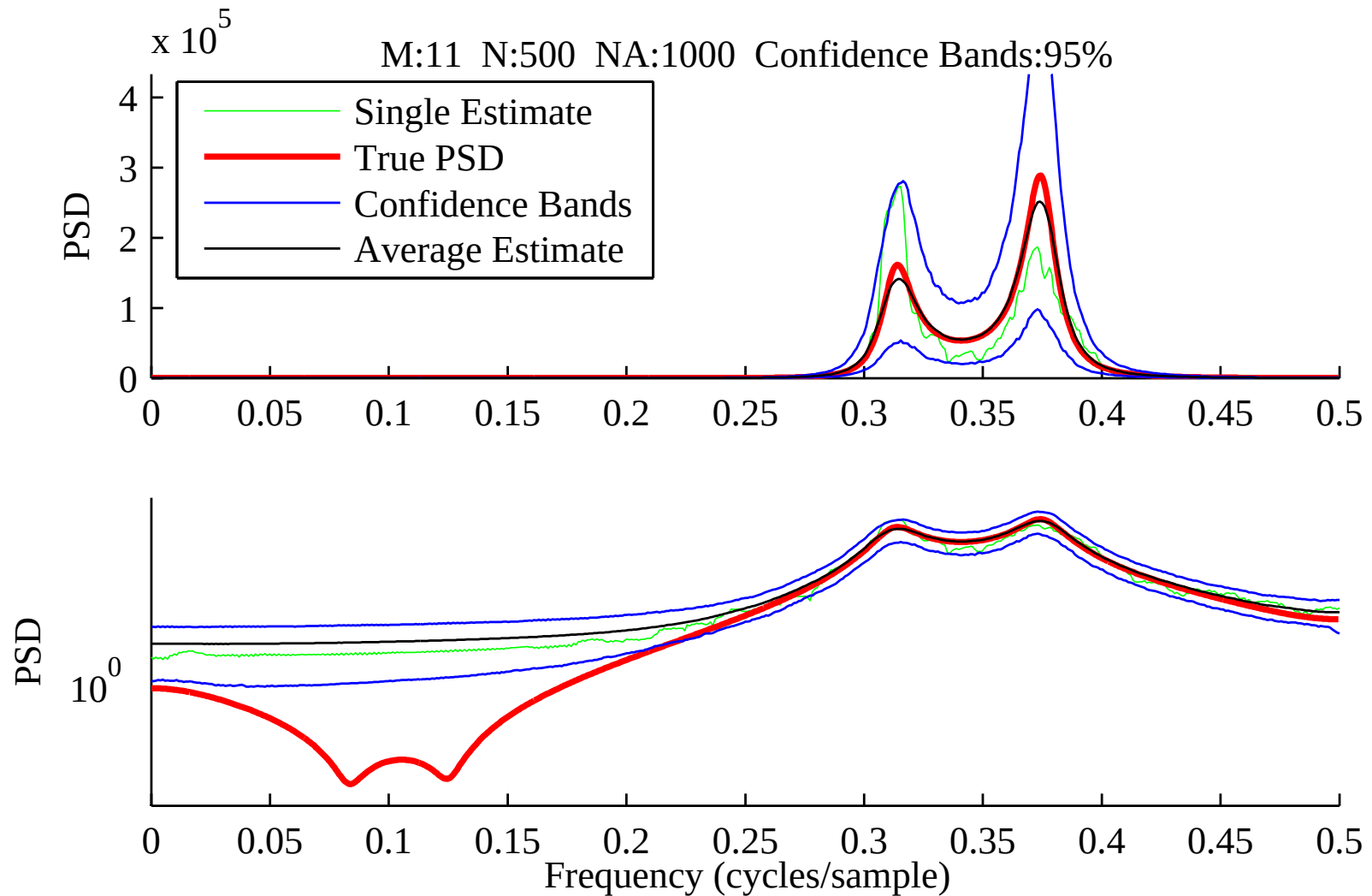
$$\hat{R}_x^{(\text{PS})}(e^{j\omega_k}) \triangleq \frac{1}{2M+1} \sum_{j=-M}^M \hat{R}_x(e^{j\omega_k-j})$$

- One approach to smoothing is to apply a moving average filter in the frequency domain
- In the definition above (borrowed from the book), only applies if $\hat{R}_x(e^{j\omega})$ is known at discrete frequencies $\omega_k \triangleq (2\pi/N)k$
- This is usually the case because we estimate using the FFT
- Since the examples at these discrete frequencies are approximately uncorrelated

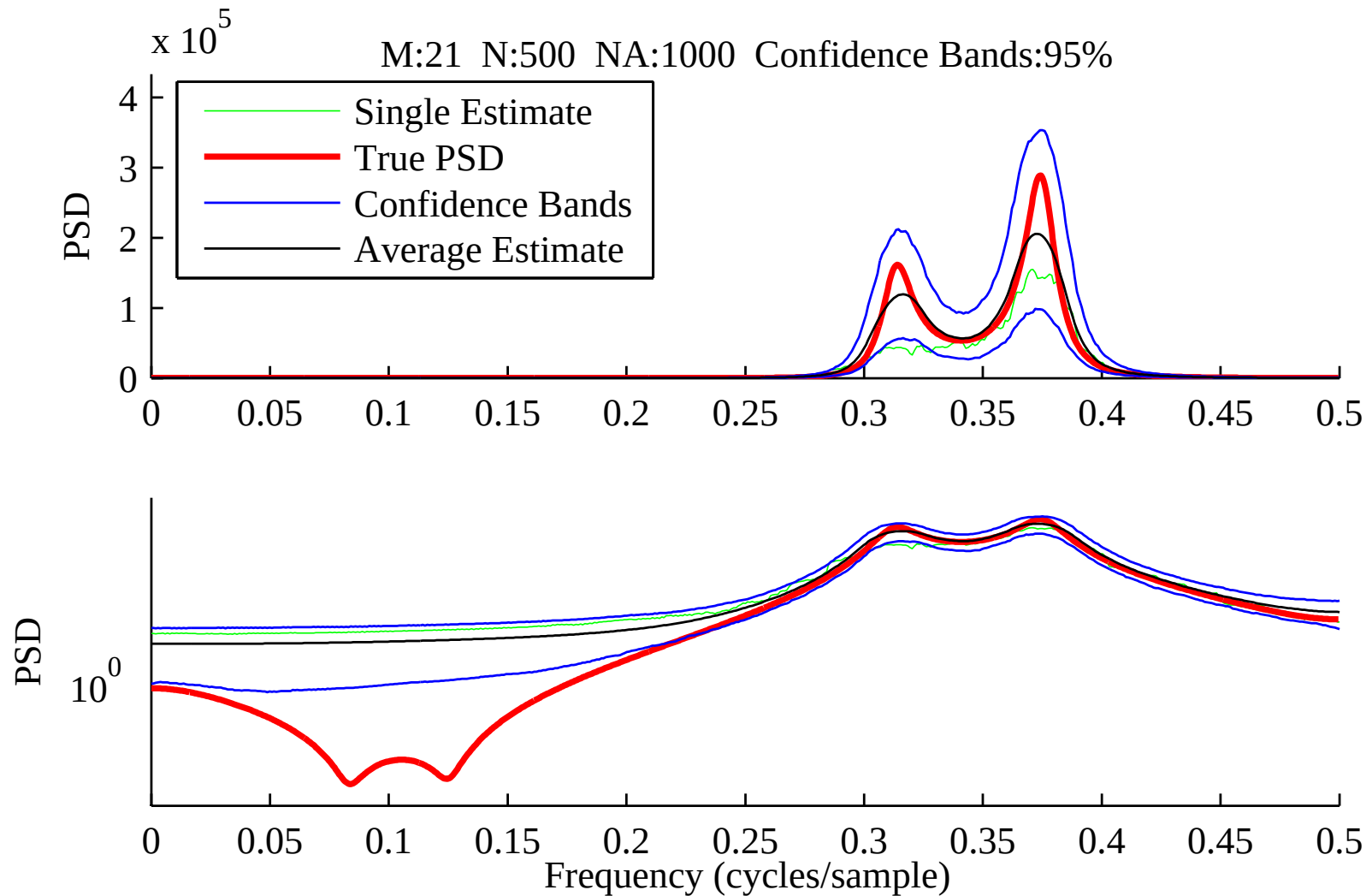
$$\text{var}\{\hat{R}_x^{(\text{PS})}(e^{j\omega_k})\} \approx \frac{1}{2M+1} \text{var}\{\hat{R}_x(e^{j\omega_k})\}$$

- However, we increase the bias, especially near sharp features in the frequency domain

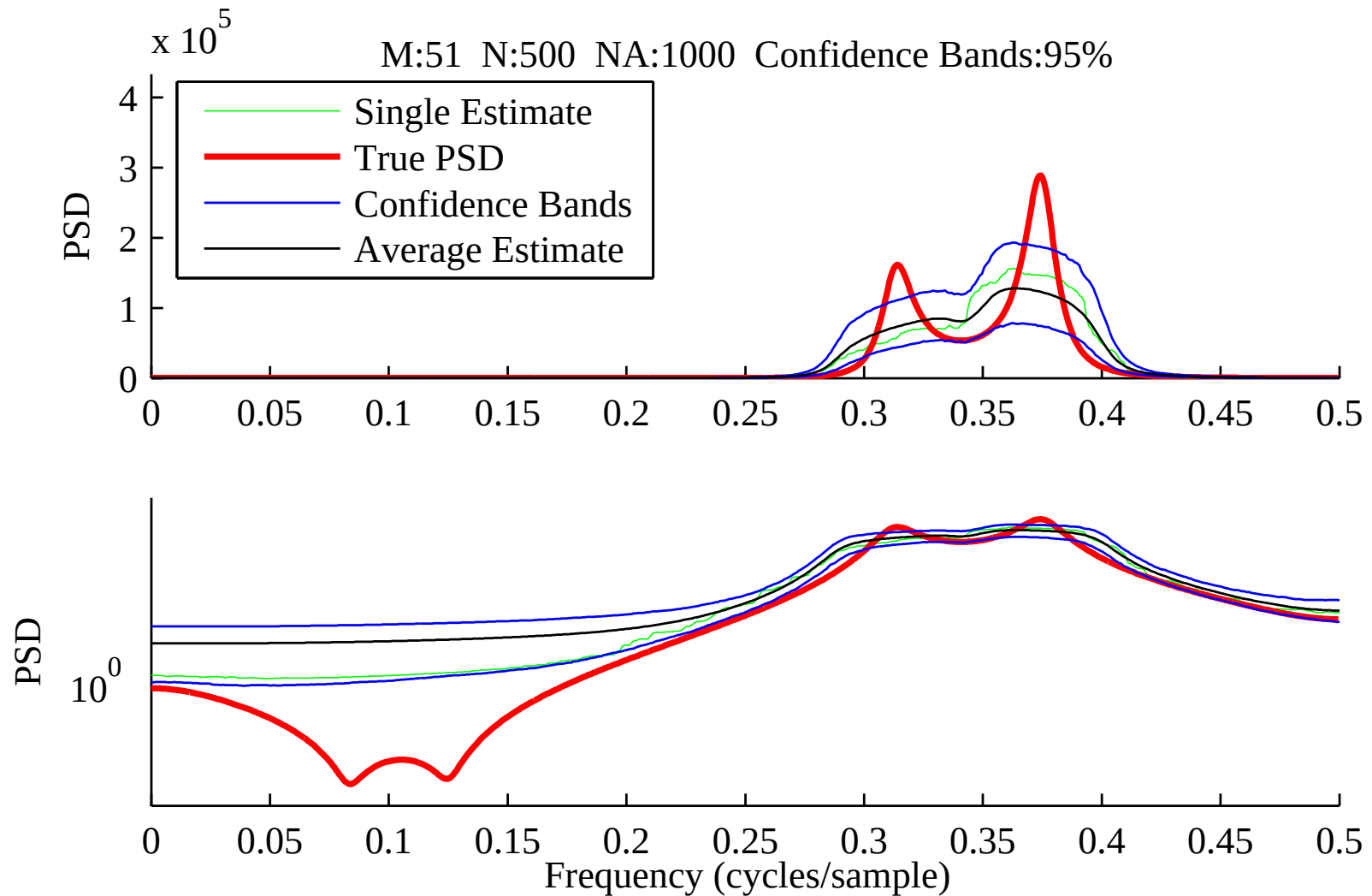
Example 5: Smoothed Periodogram Estimate



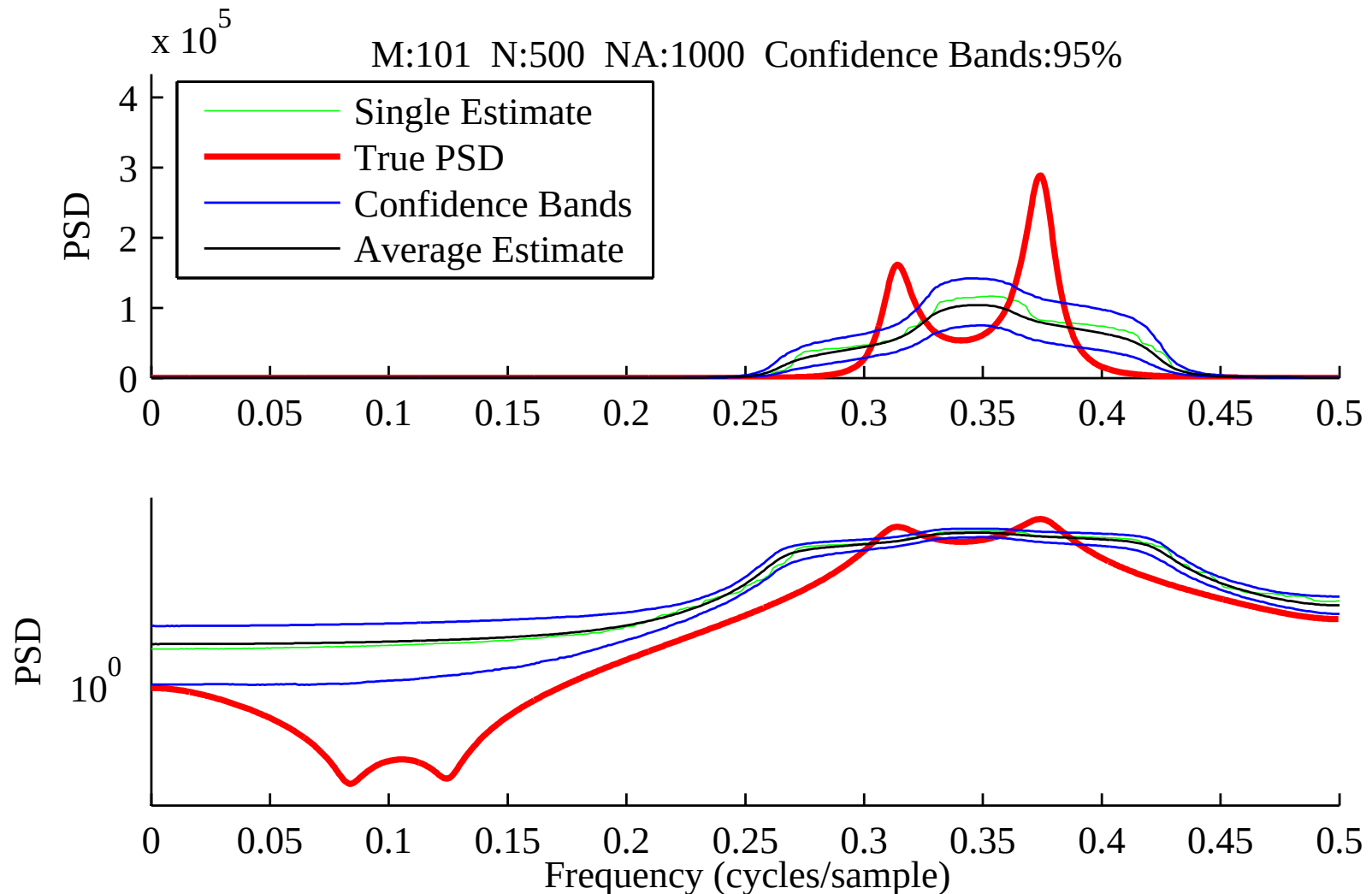
Example 5: Smoothed Periodogram Estimate



Example 5: Smoothed Periodogram Estimate



Example 5: Smoothed Periodogram Estimate



Example 5: MATLAB Code

```
%function [] = Windows;
close all;
clear all;
Q = [11,21,51,101];           % MA filter order
N = 500;                      % Length of signal segment
NA = 1000;                   % No. averages
P = 500;                     % Padding to eliminate transient
cb = 95;                     % Confidence Bands
NZ = 1024;

b = poly([-0.8,0.97*exp(j*pi/4),0.97*exp(-j*pi/4),...
          0.97*exp(j*pi/6),0.97*exp(-j*pi/6)]); % Numerator
a = poly([ 0.8,0.95*exp(j*3*pi/4),0.95*exp(-j*3*pi/4),...
          0.95*exp(j*2.5*pi/4),0.95*exp(-j*2.5*pi/4)]); % Denominator
b = b*sum(a)/sum(b);          % Set DC gain to 1

[R,w] = freqz(b,a,NZ);
R2 = abs(R).^2;
f = w/(2*pi);

%=====
% Plot Estimate
%=====
for c1 = 1:length(Q),
    nx = P+N;
    Rh = zeros(NA,NZ/2+1);
    kh = 1:NZ/2+1;
```

```

fh = (kh-1)/NZ;
for c2 = 1:NA,
    w = randn(P+N,1);
    x = filter(b,a,w); % System with known PSD
    x = x(nx-N+1:nx); % Eliminate start-up transient (make stationary)
    rh = (1/N)*abs(fft(x,NZ)).^2;

    rh = [rh;rh;rh]; % Create 3 copies to eliminate edge effects
    rhs = filter(ones(Q(c1),1)/Q(c1),1,rh);
    rhs = rhs((Q(c1)-1)/2+(NZ+1:2*NZ)); % Note shift to account for filter delay

    Rh(c2,:) = rhs(kh).';
end;

Rha = mean(Rh); % Average
Rhu = prctile(Rh,100-(100-cb)/2); % Upper confidence band
Rh1 = prctile(Rh,(100-cb)/2); % Lower confidence band

figure;
FigureSet(1,'LTX');
subplot(2,1,1);
h = plot(fh,Rh(1,:), 'g',f,R2, 'r',fh,Rh1, 'b',fh,Rhu, 'b',fh,Rha, 'k');
set(h(1), 'LineWidth',0.3);
set(h(2), 'LineWidth',1.3);
set(h(3:4), 'LineWidth',0.5);
set(h(5), 'LineWidth',0.5);
box off;
ylabel('PSD');
title(sprintf('M:%d N:%d NA:%d Confidence Bands:%d%%',Q(c1),N,NA,cb));
xlim([0 0.5]);
ylim([0 max(R2)*1.5]);

```

```

        AxisSet(8);
        legend(h([1 2 4 5]), 'Single Estimate', 'True PSD', 'Confidence Bands', 'Average Estimate', 2);
subplot(2,1,2);
h = semilogy(fh,Rh(1,:), 'g', f,R2, 'r', fh,Rh1, 'b', fh,Rhu, 'b', fh,Rha, 'k');
set(h(1)    , 'LineWidth', 0.3);
set(h(2)    , 'LineWidth', 1.3);
set(h(3:4)  , 'LineWidth', 0.5);
set(h(5)    , 'LineWidth', 0.5);
box off;
ylabel('PSD');
xlabel('Frequency (cycles/sample)');
xlim([0 0.5]);
ylim([0.2*min(R2) max(R2)*5]);
AxisSet(8);
print(sprintf('SPeriodogramAverage%04d',Q(c1)), '-depsc');
end;

```

Periodogram Smoothing Comments

$$\hat{R}_x^{(\text{PS})}(e^{j\omega_k}) \triangleq \frac{1}{2M+1} \sum_{j=-M}^M \hat{R}_x(e^{j\omega_k-j})$$

$$\text{var} \left\{ \hat{R}_x^{(\text{PS})}(e^{j\omega}) \right\} \approx \frac{1}{2M+1} \text{var} \left\{ R_x(e^{j\omega}) \right\}$$

- Smoothing the periodogram is a good idea
- Trades reduced frequency resolution and increased bias for reduced variance
- The moving average is just a lowpass filter
- There are many other lowpass filters that might perform better
- Examples: smoothing splines, local polynomial models, weighted averaging (kernel smoothing)

Blackman-Tukey Spectral Estimation Idea

$$\mathbb{E} \left[\hat{R}_x(e^{j\omega}) \right] = \frac{1}{2\pi} \int_{-\pi}^{\pi} R_x(e^{ju}) \frac{1}{N} R_w(e^{j(\omega-u)}) du$$

- Recall that windowing the signal is equivalent to convolving the signal's PSD with the window's PSD, in expectation
- Despite the blurring and smoothing of this effect, the estimate is still has too high of variance
- We can increase the smoothing and reduce the variance by multiplying the autocorrelation by an even shorter window

Blackman-Tukey Spectral Estimation

$$\hat{R}_x^{(\text{BT})}(e^{j\omega}) \triangleq \sum_{\ell=-(L-1)}^{L-1} \hat{r}_x(\ell) w_a(\ell) e^{-j\omega\ell}$$

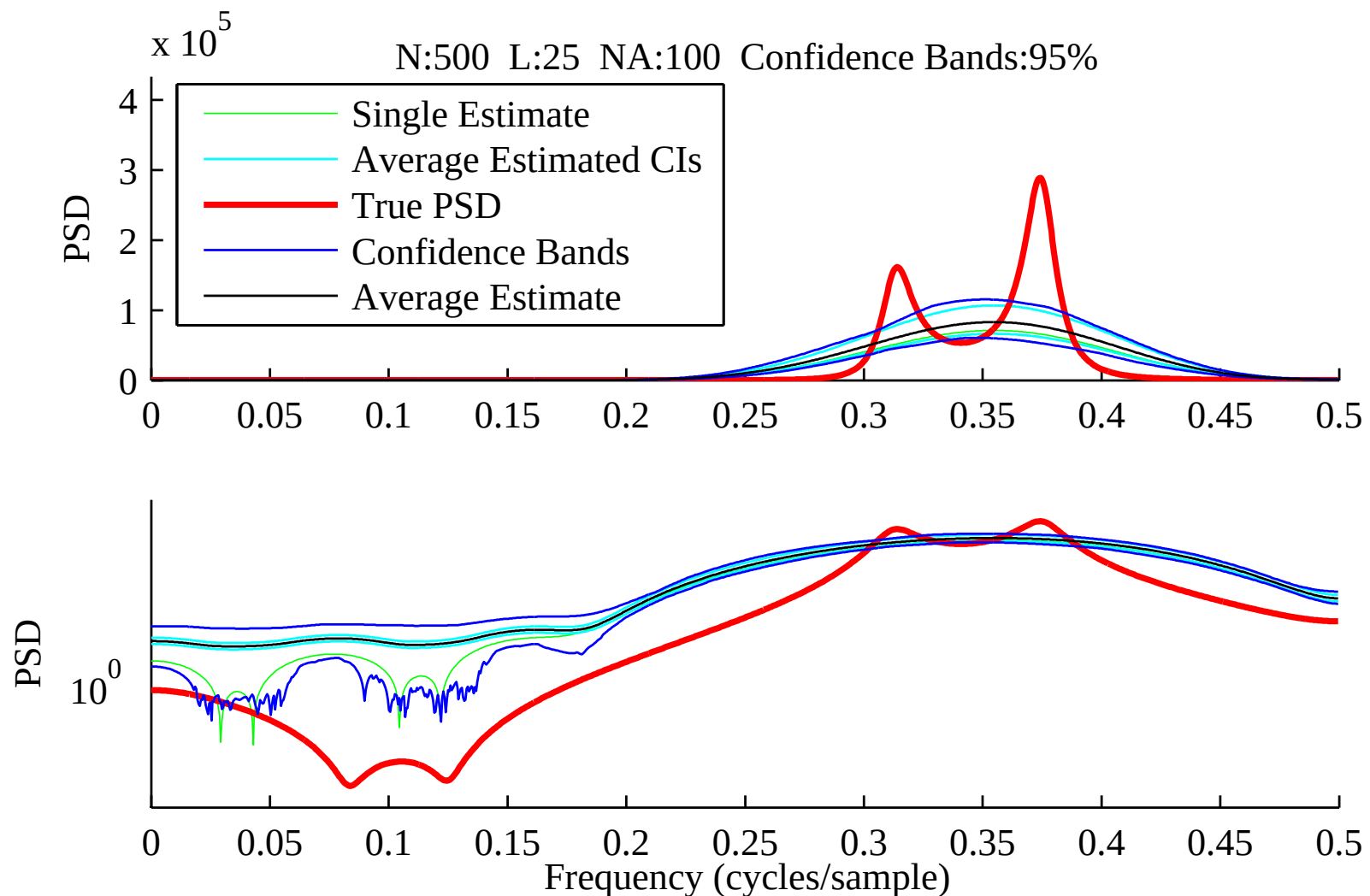
- Here the window is user specified and has a finite duration of $2L - 1$ samples
- Called the **correlation** or **lag window**
- In order for the estimate to be nonnegative, $W(e^{j\omega}) \geq 0$ for all ω
- Not all of the windows we described have this property
- In the frequency domain, the product $\hat{r}_x(\ell)w(\ell)$ is equivalent to convolving with $W(e^{j\omega})$
- Assuming the window's spectrum is bumped shape, this convolution smooths the spectrum
- Note that the FFT can be used to calculate both $\hat{r}_x(\ell)$ and $\hat{R}_x^{(\text{BT})}(e^{j\omega})$ efficiently

Blackman-Tukey Bias & Variance

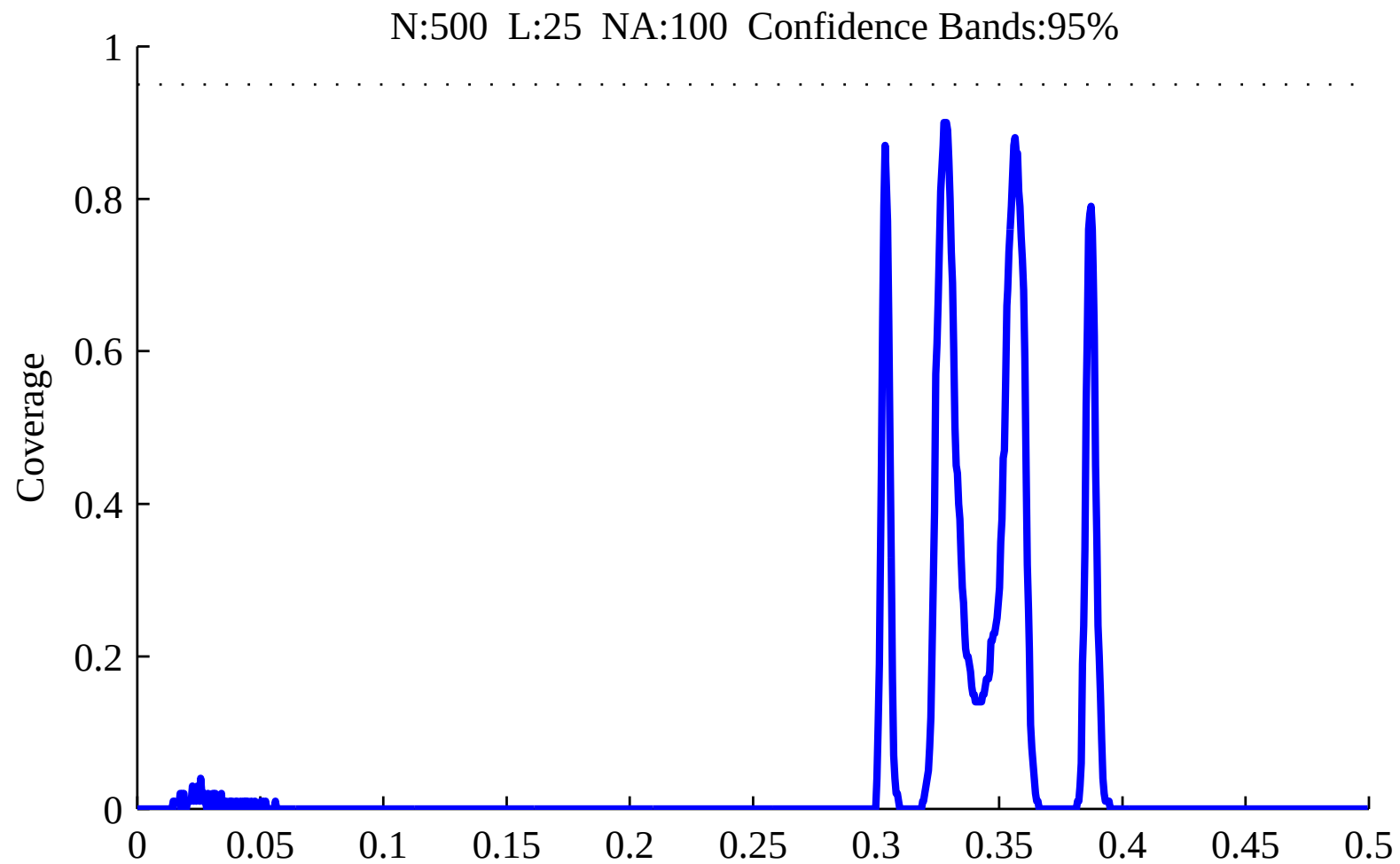
$$\hat{R}_x^{(\text{BT})}(e^{j\omega}) \triangleq \sum_{\ell=-(L-1)}^{L-1} \hat{r}_x(\ell) w_a(\ell) e^{-j\omega\ell}$$

- Conceptually, this is using an even more biased estimate of the autocorrelation: $\hat{r}_x(\ell) w_a(\ell)$
- The additional bias is towards zero
- Thus, it reduces the variance of both $\hat{r}_x(\ell)$ and $\hat{R}_x^{(\text{BT})}(e^{j\omega})$
- The user controls this tradeoff by picking the **correlation window** $w_a(\ell)$
- The window can be any *positive definite* window with even symmetry (why?)
- The length of the window L has a larger impact than the shape

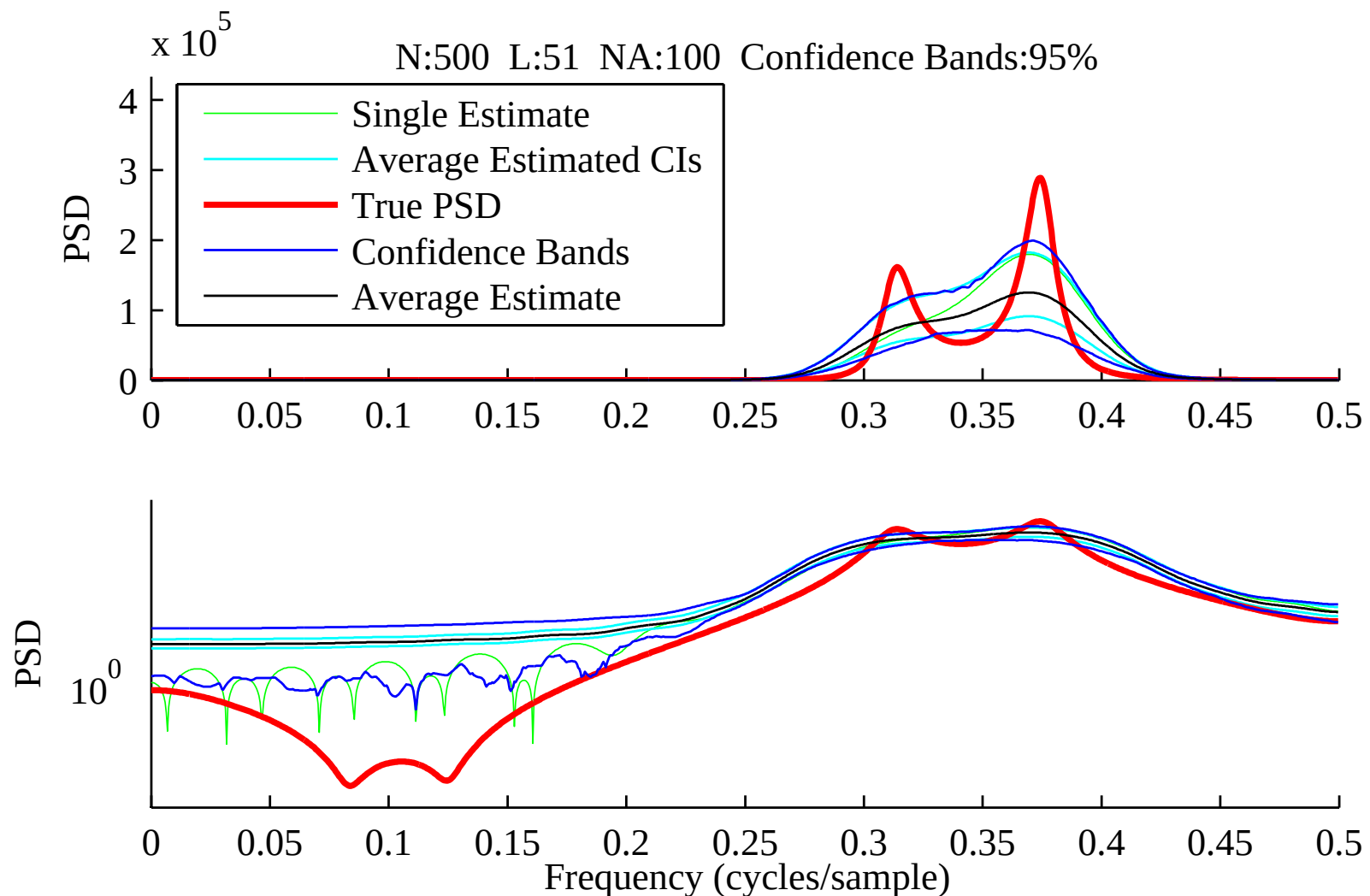
Example 6: Blackman-Tukey Estimate



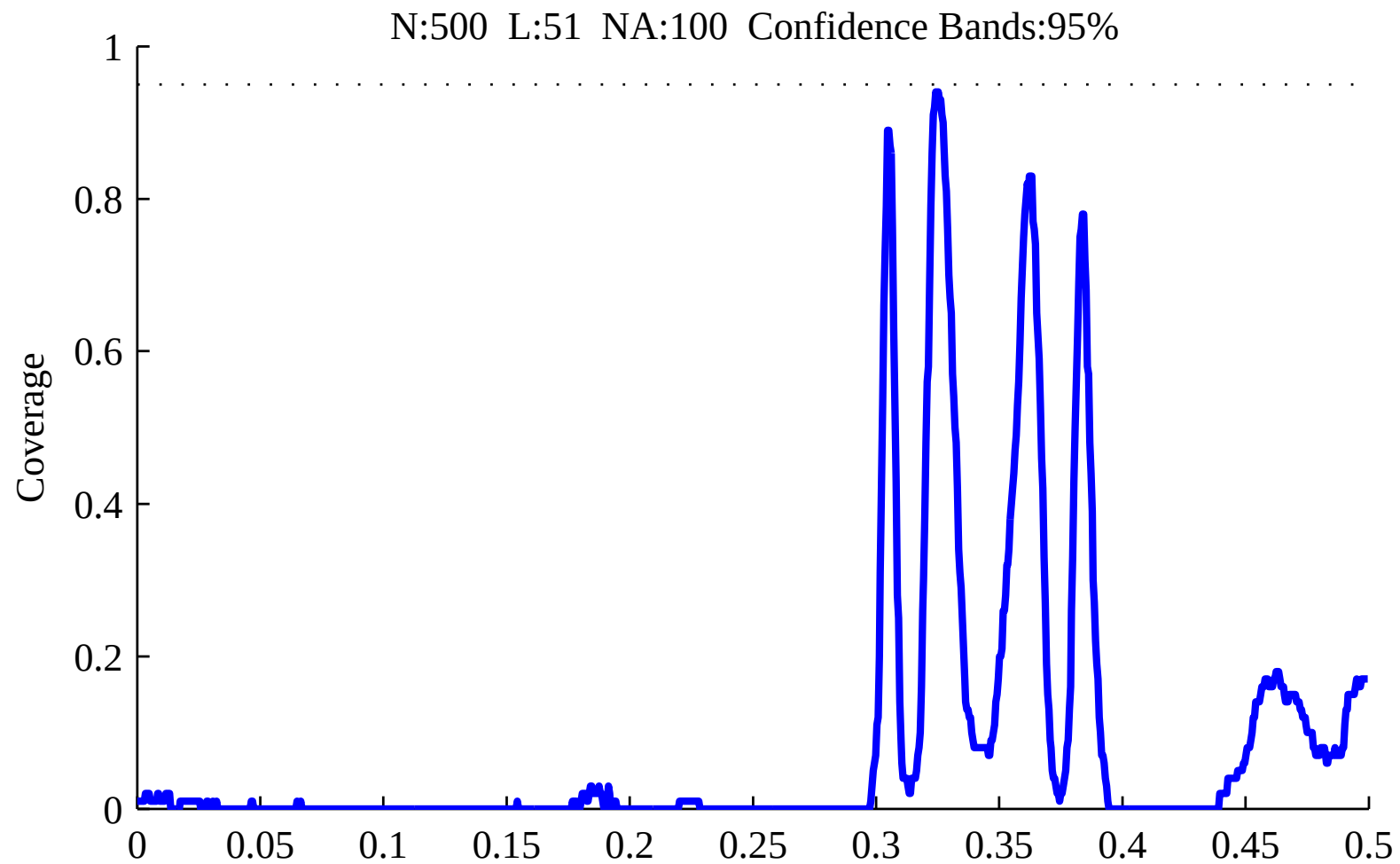
Example 7: Blackman-Tukey Coverage



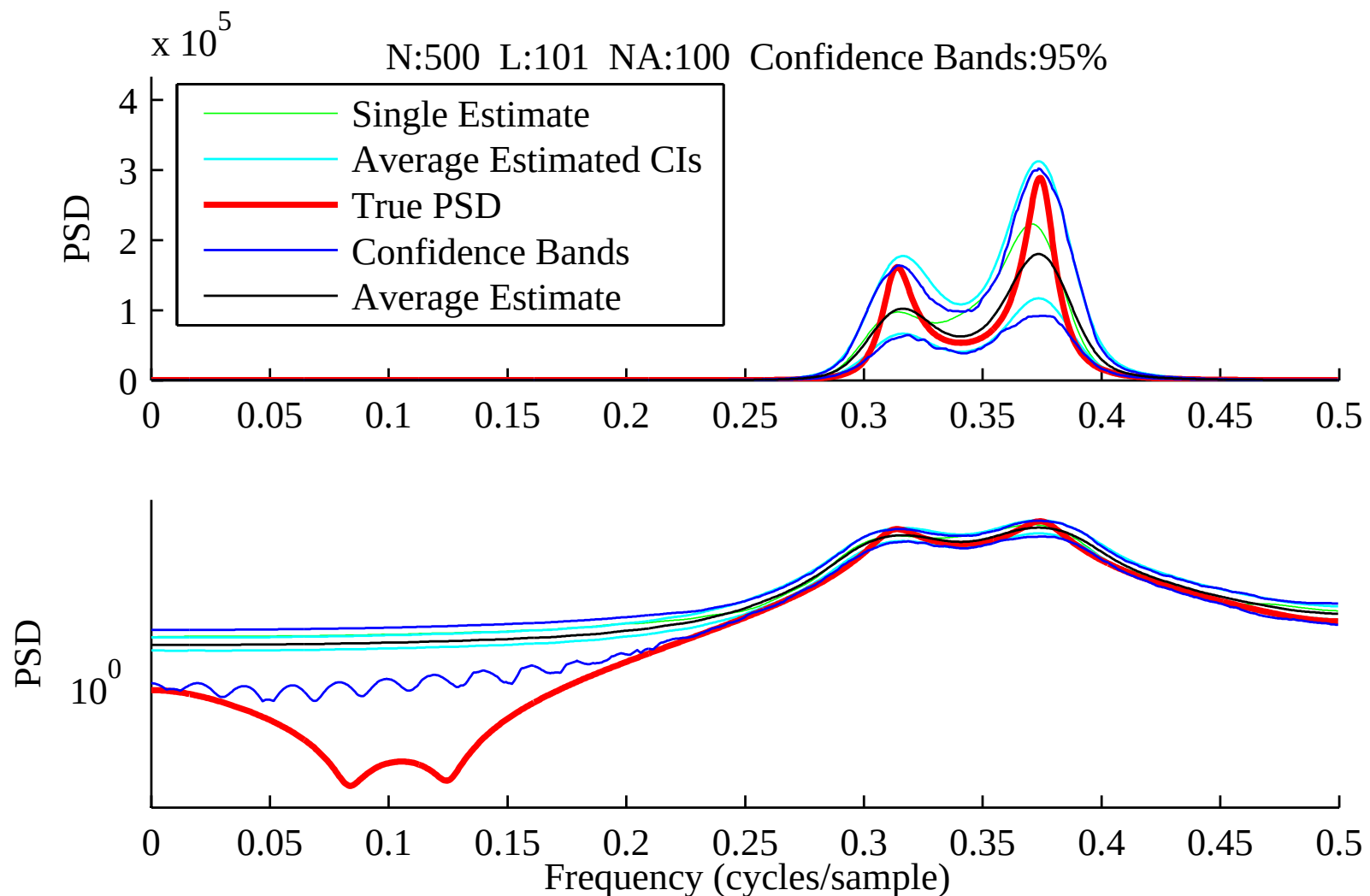
Example 7: Blackman-Tukey Estimate



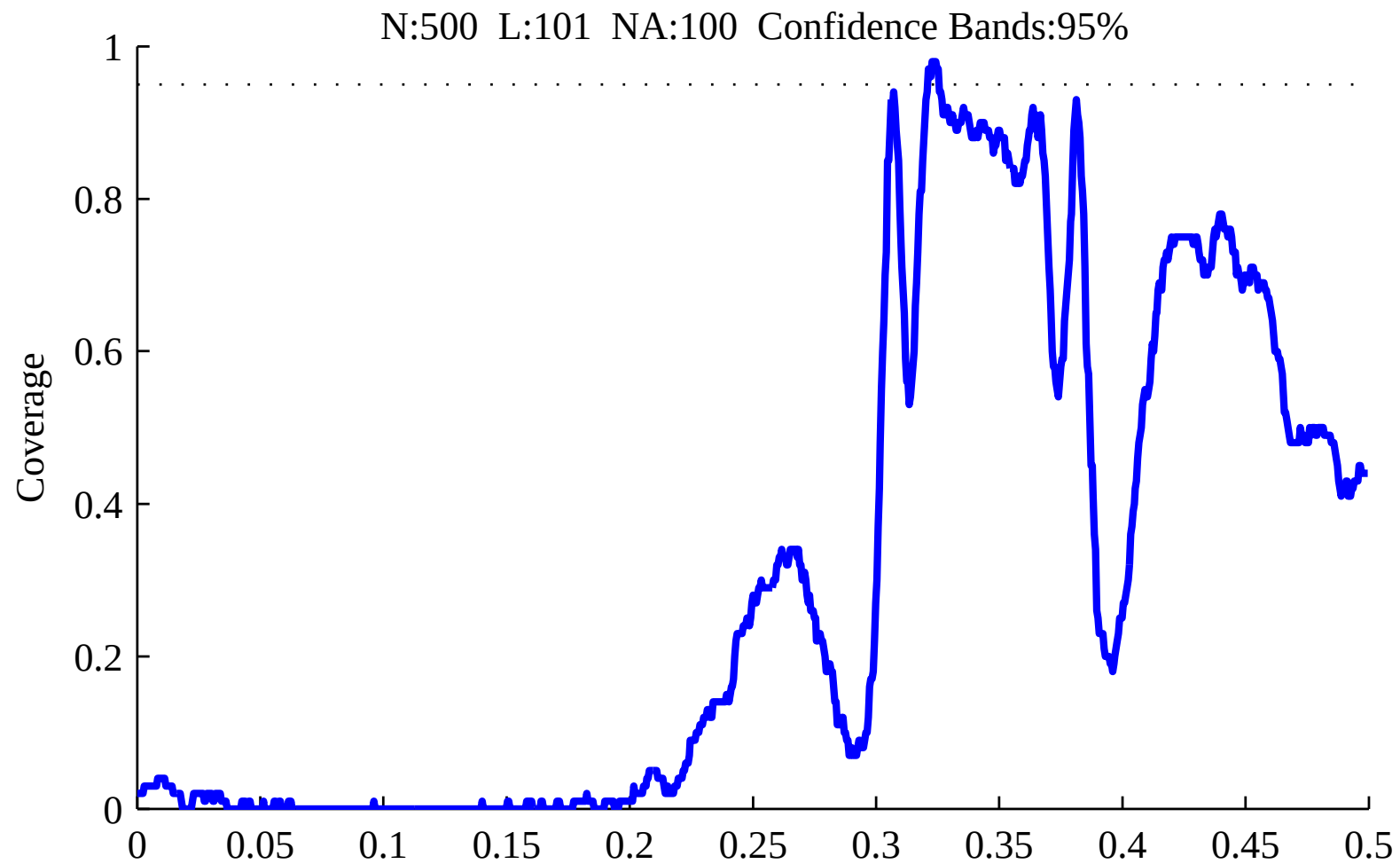
Example 8: Blackman-Tukey Coverage



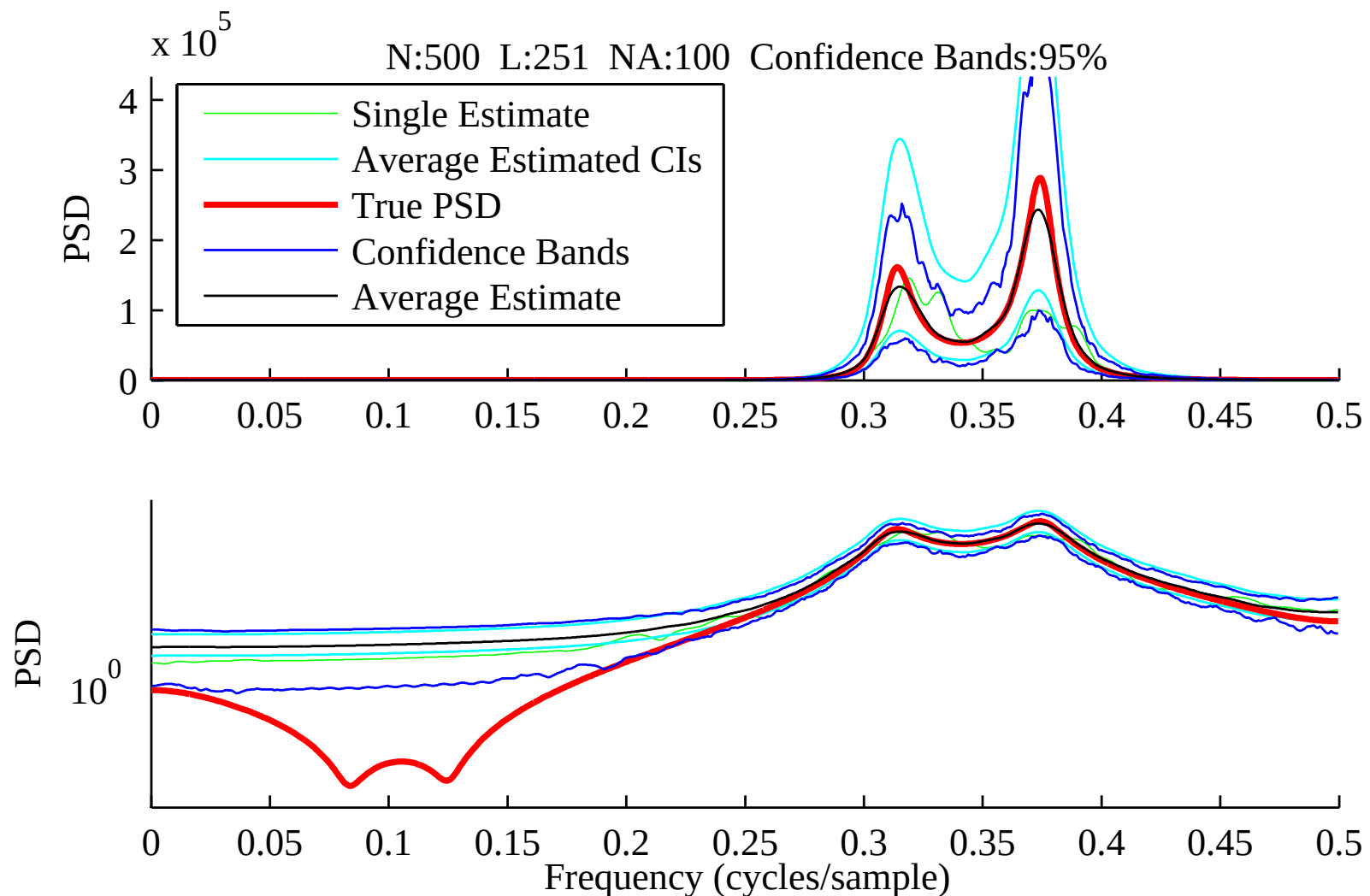
Example 8: Blackman-Tukey Estimate



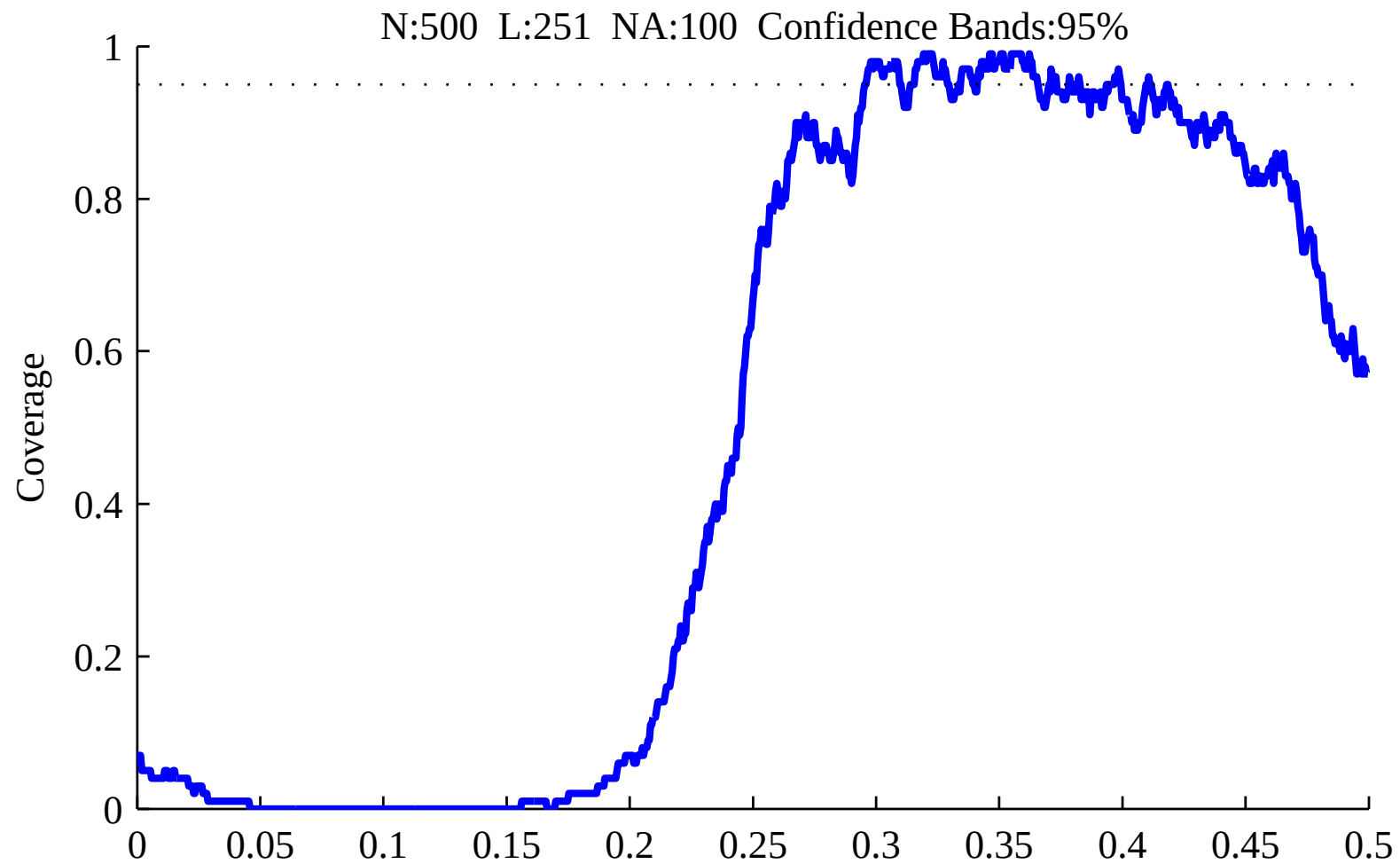
Example 9: Blackman-Tukey Coverage



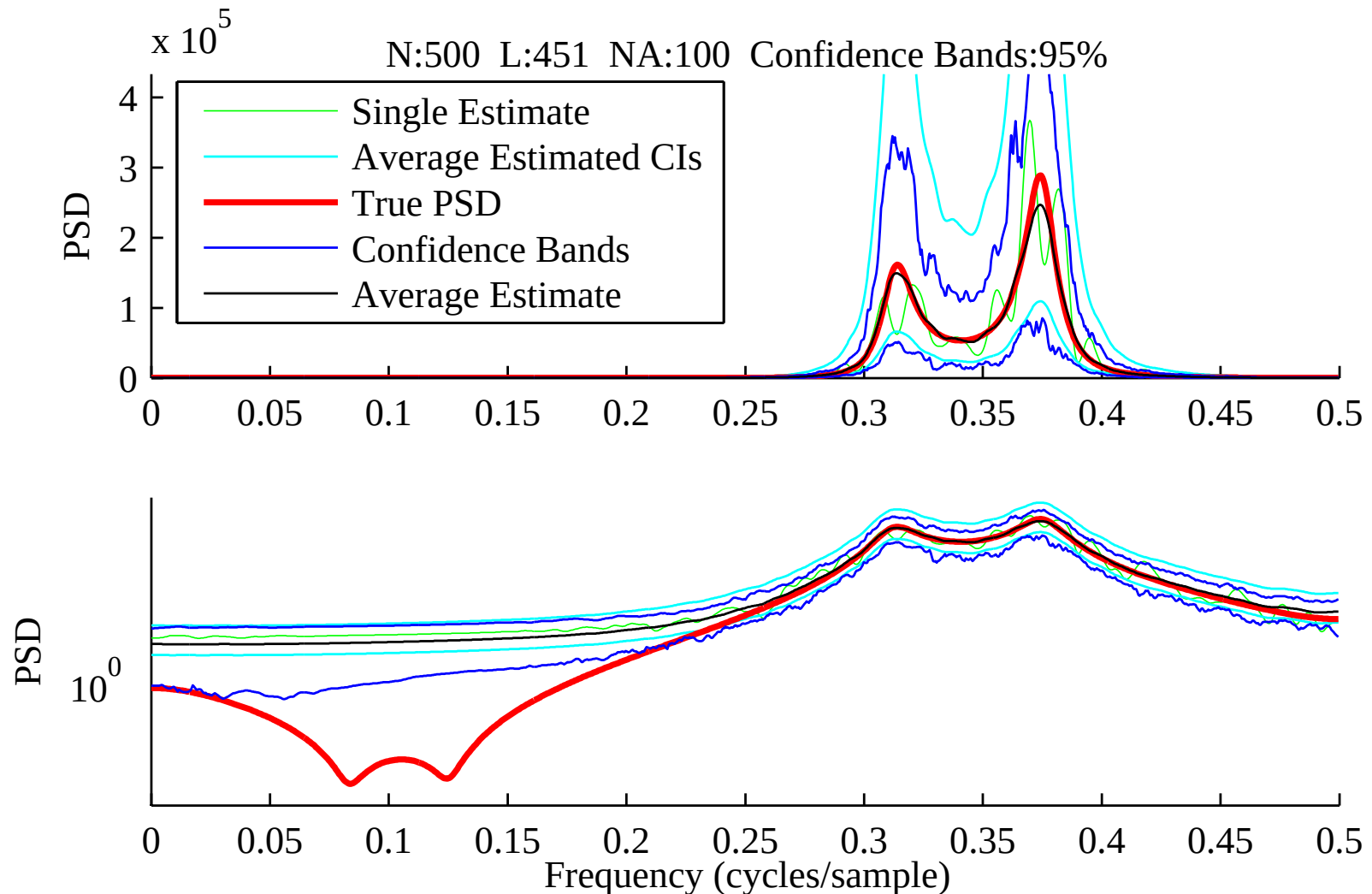
Example 9: Blackman-Tukey Estimate



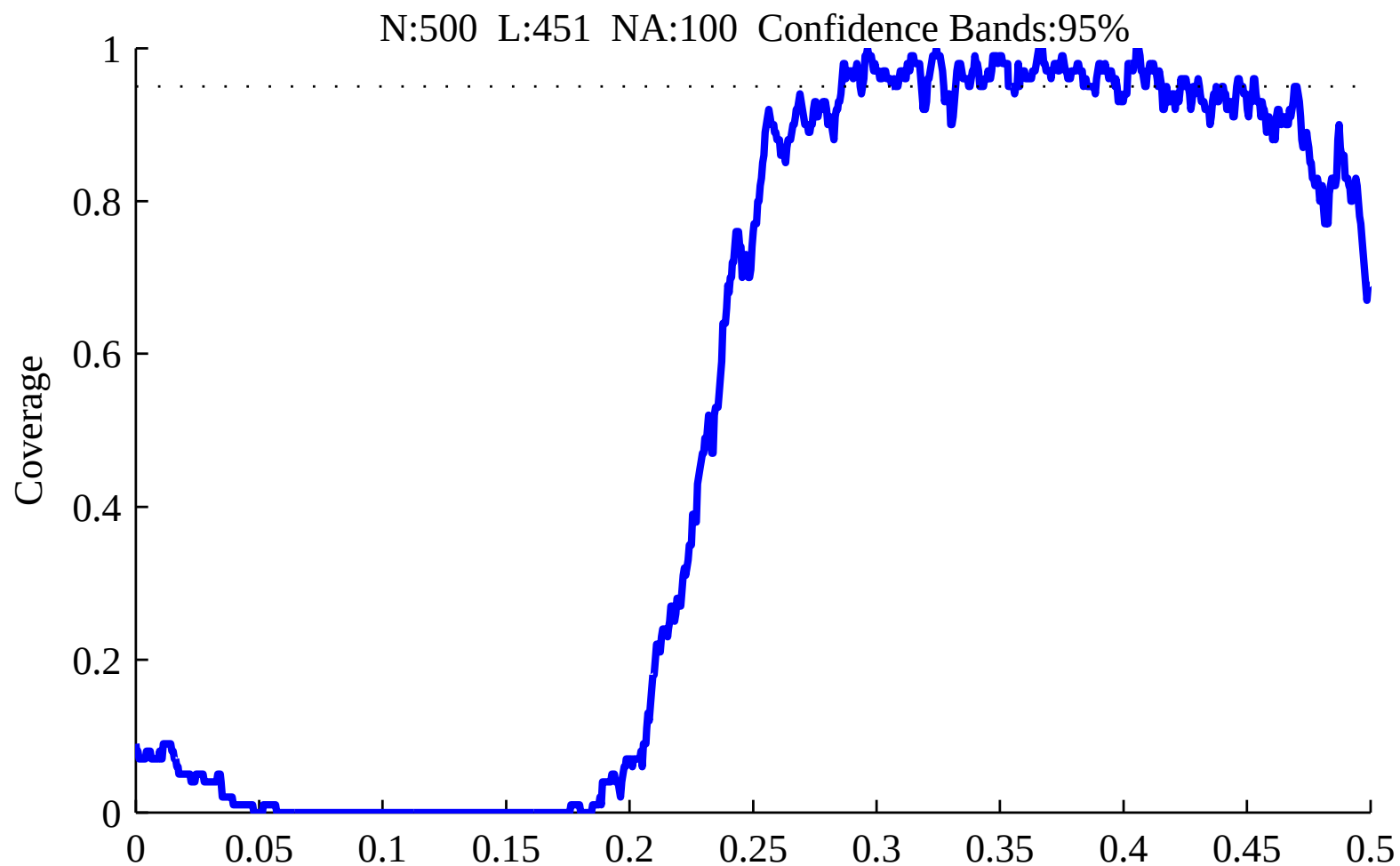
Example 10: Blackman-Tukey Coverage



Example 10: Blackman-Tukey Estimate



Example 11: Blackman-Tukey Coverage



Example 11: MATLAB Code

```
%function [] = BlackmanTukeyEx();
close all;
clear all;

L = [25,51,101,201,451]; % MA filter order
N = 500;                  % Length of signal segment
NA = 2000;                % No. averages
P = 500;                  % Padding to eliminate transient
cb = 95;                  % Confidence Bands
NZ = 2^9;                 % No. zeros to pad
np = 2^nextpow2(2*N-1);   % Figure out how much to pad the signal

al = 1-cb/100;            % Alpha

b = poly([-0.8,0.97*exp(j *pi/4),0.97*exp(-j *pi/4),...
          0.97*exp(j *pi/6),0.97*exp(-j *pi/6)]); % Numerator coefficients
a = poly([ 0.8,0.95*exp(j*3*pi/4),0.95*exp(-j*3*pi/4),...
          0.95*exp(j*2.5*pi/4),0.95*exp(-j*2.5*pi/4)]); % Denominator coefficients
b = b*sum(a)/sum(b);      % Scale DC gain to 1

[R,w] = freqz(b,a,NZ);
R      = abs(R).^2;
R      = R';
f      = w/(2*pi);

nL     = length(L);
```

```

%=====
% Plot Estimate
%=====
Bias = zeros(nL,NZ);
Var  = zeros(nL,NZ);

for c1 = 1:nL,
    nx = P+N;
    Rh = zeros(NA,NZ);
    Rhu = zeros(NA,NZ);
    Rhl = zeros(NA,NZ);
    Rhc = zeros(NA,NZ);
    kh = 1:NZ;
    %fh = (kh-1)/(2*NZ);

    no = (L(c1)+1)/2;          % Number of one-sided points to included in estimate
    wn = blackman(no*2-1);     % Create window
    wn = wn/max(wn);           % Make maximum = 1
    v = 2*N/(sum(wn.^2));

    for c2 = 1:NA,
        w = randn(P+N,1);
        x = filter(b,a,w); % System with known PSD
        x = x(nx-N+1:nx); % Eliminate start-up transient (make stationary)

        xf = fft(x,np);          % Calculate FFT of x
        r = ifft(xf.*conj(xf)); % Calculate biased autocorrelation
        r = real(r(1:nx))/N;     % Eliminate superfluous zeros and nearly zero imaginary part
        r = [r(no:-1:2);r(1:no)]; % Make two-sided
        rh = abs(fft(r.*wn,2*NZ)); % Window autocorrelation and calculate the estimate
    end
end

```

```

    rh1 = exp(log(rh) - log(chi2inv(1-al/2,v)/v));
    rhu = exp(log(rh) + log(v/chi2inv(al/2,v) ));

    Rh (c2,:) = rh(kh).';
    Rh1(c2,:) = rh1(kh).';
    Rhu(c2,:) = rhu(kh).';
    Rhc(c2,:) = rh1(kh)<R' & R'<rhu(kh);
    end;

    Rha = mean(Rh); % Average
    Rhle = mean(Rh1); % Average lower confidence interval
    Rhue = mean(Rhu); % Average upper confidence interval
    Rhu = prctile(Rh,100-(100-cb)/2); % Upper confidence band
    Rh1 = prctile(Rh, (100-cb)/2); % Lower confidence band

    Bias(c1,:) = Rha-R;
    Var (c1,:) = mean((Rh - ones(NA,1)*Rha).^2);
    MSE (c1,:) = mean((Rh - ones(NA,1)*R ).^2);

    if 0,
        figure;
        FigureSet(1,'LTX');
        subplot(2,1,1);
        h = plot(f,Rh(1,:),'g',f,Rhue,'c',f,Rhle,'c',f,R,'r',f,Rh1,'b',f,Rhu,'b',f,Rha,'k');
        set(h(1) , 'LineWidth',0.3);
        set(h(2:3),'LineWidth',0.5);
        set(h(4) , 'LineWidth',1.3);
        set(h(5:6),'LineWidth',0.5);
        set(h(7) , 'LineWidth',0.5);
        box off;
    end

```

```

        ylabel('PSD');
        title(sprintf('N:%d L:%d NA:%d Confidence Bands:%d%%',N,L(c1),NA,cb));
        xlim([0 0.5]);
        ylim([0 max(R)*1.5]);
        AxisSet(8);
        legend(h([1 2 4 5 7]),'Single Estimate','Average Estimated CIs','True PSD','Confidence Bands',
subplot(2,1,2);
        h = semilogy(f,Rh(1,:), 'g',f,Rhue, 'c',f,Rhle, 'c',f,R, 'r',f,Rhl, 'b',f,Rhu, 'b',f,Rha, 'k');
        set(h(1), 'LineWidth',0.3);
        set(h(2:3), 'LineWidth',0.5);
        set(h(4), 'LineWidth',1.3);
        set(h(5:6), 'LineWidth',0.5);
        set(h(7), 'LineWidth',0.5);
        box off;
        ylabel('PSD');
        xlabel('Frequency (cycles/sample)');
        xlim([0 0.5]);
        ylim([0.2*min(R) max(R)*5]);
        AxisSet(8);
        print(sprintf('BlackmanTukeyAverage%04d',L(c1)),'-depsc');

figure;
FigureSet(2,'LTX');
h = plot(f,mean(Rhc),[0 0.5],[0.95 0.95], 'k:');
set(h(1), 'LineWidth',1.5);
box off;
ylabel('Coverage');
title(sprintf('N:%d L:%d NA:%d Confidence Bands:%d%%',N,L(c1),NA,cb));
xlim([0 0.5]);
ylim([0 1]);

```

```
AxisSet(8);  
%legend(h([1 2 4 5 7]),'Single Estimate','Average Estimated Confidence Bands','True PSD','Confidence  
print(sprintf('BlackmanTukeyConfidenceCoverage%04d',L(c1)),'-depsc');  
end;  
  
drawnow;  
end;
```

Blackman-Tukey Mean

$$\begin{aligned}\hat{R}_x^{(\text{BT})}(e^{j\omega}) &\triangleq \sum_{\ell=-(L-1)}^{L-1} \hat{r}_x(\ell) w_a(\ell) e^{-j\omega\ell} \\ \text{E} \left[\hat{R}_x^{(\text{BT})}(e^{j\omega}) \right] &= \sum_{\ell=-(L-1)}^{L-1} \text{E} [\hat{r}_x(\ell)] w_a(\ell) e^{-j\omega\ell} \\ &= \sum_{\ell=-(L-1)}^{L-1} \left[r_x(\ell) \frac{1}{N} r_w(\ell) \right] w_a(\ell) e^{-j\omega\ell} \\ &= R_x(e^{j\omega}) * W(e^{j\omega}) * W_a(e^{j\omega})\end{aligned}$$

- Recall that the periodogram had an expected value given by the convolution of the Bartlett window's spectrum and the true PSD
- Multiplication by another window causes another convolution
- Text assumes the data window $w(\ell)$ is rectangular

Blackman-Tukey Mean

$$\mathbb{E} \left[\hat{R}_x^{(\text{BT})}(e^{j\omega}) \right] = R_x(e^{j\omega}) * W(e^{j\omega}) * W_a(e^{j\omega})$$

- Note that the convolution is circular since each term is a periodic function of ω
- The estimate is asymptotically unbiased if $w_a(0) = 1$
- As L and N approach ∞ , both windows become impulse trains

Blackman-Tukey Sampling Distribution

$$\hat{R}_x^{(\text{BT})}(e^{j\omega}) \triangleq \sum_{\ell=-(L-1)}^{L-1} \hat{r}_x(\ell) w_a(\ell) e^{-j\omega\ell}$$

- Sampling properties of the spectral estimates are extremely complicated
- Exact results can only be obtained in special circumstances
- Most expressions for mean, variance, covariance, and distributions are asymptotic and only apply when N is large
- The asymptotic expressions assume that L increases with N , but more slowly than N : $L \rightarrow \infty$, $N \rightarrow \infty$, $L/N \rightarrow 0$

Blackman-Tukey Covariance

$$\begin{aligned} \text{cov}\{\hat{R}_x^{(\text{BT})}(e^{j\omega_1}), \hat{R}_x^{(\text{BT})}(e^{j\omega_2})\} \\ \approx \frac{1}{2\pi N} \int_{-\pi}^{\pi} R_x^2(e^{ju}) W_a(e^{j(\omega_1-u)}) W_a(e^{j(\omega_2-u)}) du \end{aligned}$$

- Assumptions
 - N is large so that $W_B(e^{j\omega})$ behaves like an impulse train
 - L is sufficiently large that $W_a(e^{j\omega})$ is narrow and the product $W_a(e^{j\omega_1})W_a(e^{j\omega_2})$ is negligible
- Thus, covariance increases as
 - The window's spectrum, $W_a(e^{j(\omega_1-u)})$, grows wider
 - The overlap between windows centered at different frequencies, ω_1 and ω_2 , increases
 - For estimates to be uncorrelated, they must be separated in frequency by an amount greater than the “bandwidth” of the window

Blackman-Tukey Variance

$$\text{var}\{\hat{R}_x^{(\text{BT})}(e^{j\omega_1})\} \approx \frac{1}{2\pi N} \int_{-\pi}^{\pi} R_x^2(e^{ju}) W_a^2(e^{j(\omega-u)}) du$$

If $R_x^2(e^{j\omega})$ is smooth over the width of the window and can be approximated as a constant, then

$$\text{var}\{\hat{R}_x^{(\text{BT})}(e^{j\omega})\} \approx R_x^2(e^{j\omega}) \frac{1}{2\pi N} \int_{-\pi}^{\pi} W_a^2(e^{j(\omega-u)}) du$$

$$= R_x^2(e^{j\omega}) \frac{E_w}{N} \quad 0 < \omega < \pi$$

$$E_w = \sum_{\ell=-(L-1)}^{L-1} w_a^2(\ell)$$

- E_w is the energy of the correlation window
- $\frac{E_w}{N}$ is called the **variance reduction factor** or **variance ratio**

Blackman-Tukey Confidence Intervals

$$10 \log \hat{R}_x^{(\text{BT})}(e^{j\omega}) - 10 \log \frac{\chi_\nu^2(1 - \alpha/2)}{\nu} < 10 \log R_x^{(\text{BT})}(e^{j\omega}),$$

$$10 \log R_x^{(\text{BT})}(e^{j\omega}) < 10 \log \hat{R}_x^{(\text{BT})}(e^{j\omega}) + 10 \log \frac{\nu}{\chi_\nu^2(1 - \alpha/2)}$$

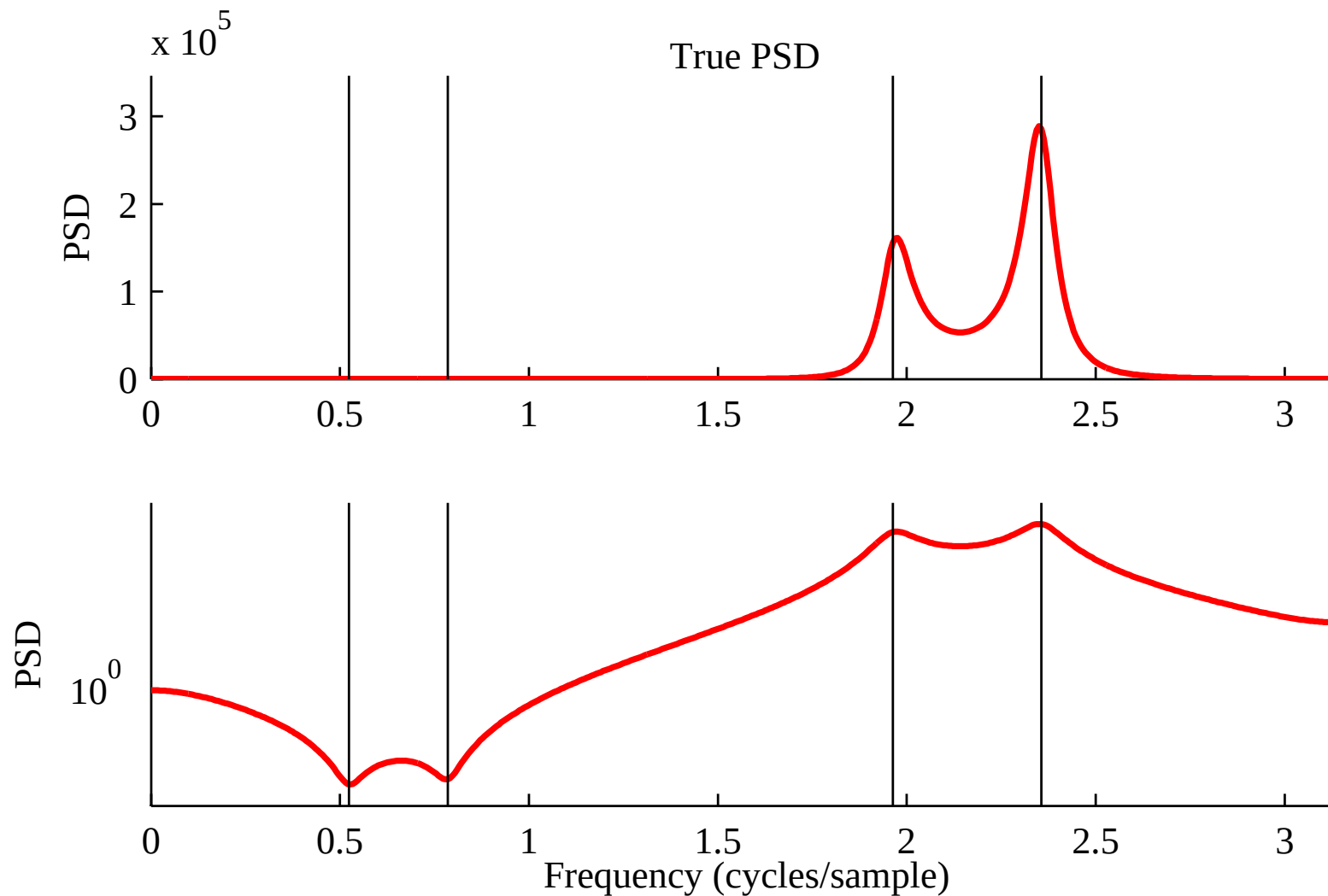
$$\nu = \frac{2N}{\sum_{\ell=-(L-1)}^L w_a^2(\ell)}$$

- These are only *approximate confidence intervals*
- Only works when
 - Bias is small $N \rightarrow \infty$
 - $L \ll N$ such that $L/N \rightarrow 0$ (this is why we don't obtain the Periodogram estimate when $L = N$)
 - This estimate is a linear combination of the periodogram estimate, so it is also approximately has a χ^2 distribution, but with more degrees of freedom

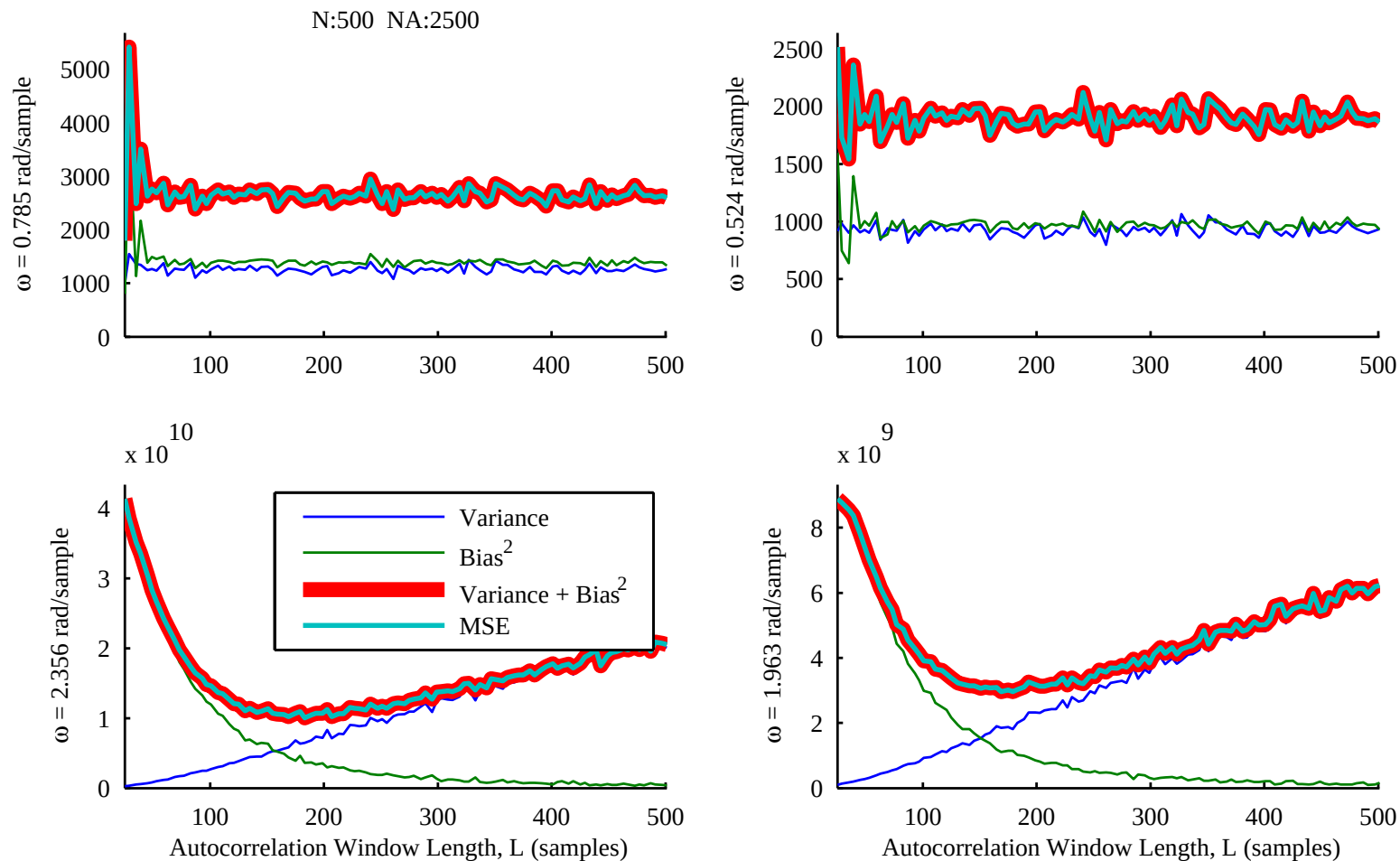
Example 12: Blackman-Tukey Bias-Variance Tradeoff

Plot the bias and variance at the frequencies of the poles and zeros of the LTI system for a range of window lengths ranging from 25–501 samples.

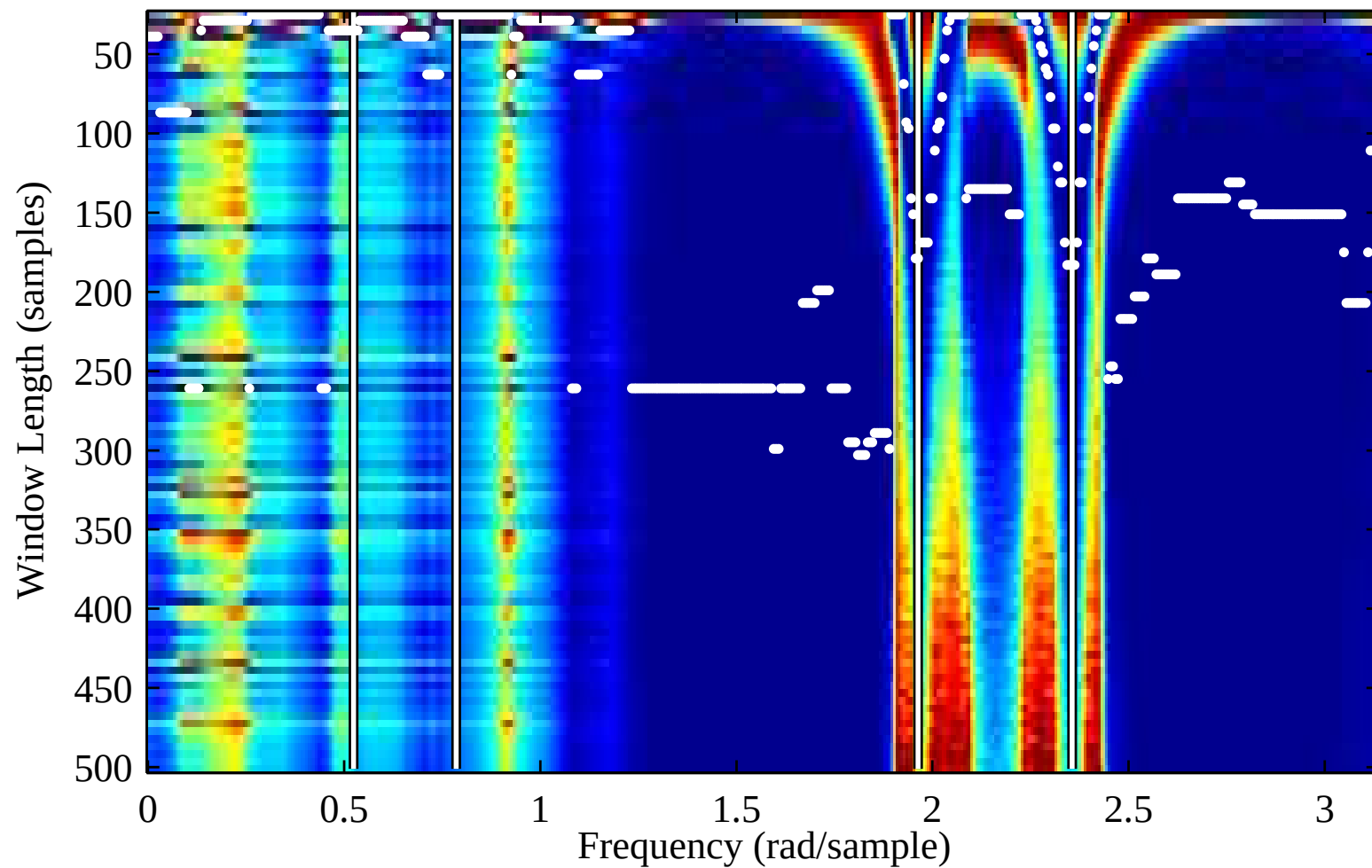
Example 12: Blackman-Tukey Bias-Variance Frequencies



Example 12: Blackman-Tukey Bias-Variance Tradeoff



Example 12: Blackman-Tukey Bias-Variance Tradeoff



Example 12: MATLAB Code

```
%function [] = BlackmanTukeyEx();
close all;
clear all;

L = round(linspace(12,250,100))*2+1; % Autocorrelation window length
ef = [pi/4 pi/6 3*pi/4 2.5*pi/4]; % Evaluation frequencies

N = 500; % Length of signal segment
NA = 2500; % No. averages
P = 500; % Padding to eliminate transient
NZ = 2^9; % No. zeros
np = 2^nextpow2(2*N-1); % Figure out how much to pad the signal

b = poly([-0.8,0.97*exp(j *pi/4),0.97*exp(-j *pi/4),...
          0.97*exp(j *pi/6),0.97*exp(-j *pi/6)]); % Numerator coefficients
a = poly([ 0.8,0.95*exp(j*3*pi/4),0.95*exp(-j*3*pi/4),...
          0.95*exp(j*2.5*pi/4),0.95*exp(-j*2.5*pi/4)]); % Denominator coefficients
b = b*sum(a)/sum(b); % Scale DC gain to 1

[R,w] = freqz(b,a,NZ);
R = abs(R).^2;
R = R';
f = w;

nL = length(L);

%=====
```

```

% Main Routine
%=====
Bias = zeros(nL,NZ);
Var  = zeros(nL,NZ);
kh   = 1:NZ;
nx   = P+N;

for c1 = 1:nL,
    Rh = zeros(NA,NZ);
    no = (L(c1)+1)/2;           % Number of one-sided points to included in estimate
    wn = blackman(no*2-1+2);    % Create window
    wn = wn(2:end-1);          % Trim MATLAB's zeros
    wn = wn/max(wn);           % Make maximum = 1

    for c2 = 1:NA,
        w = randn(P+N,1);
        x = filter(b,a,w); % System with known PSD
        x = x(nx-N+1:nx); % Eliminate start-up transient (make stationary)
        xf = fft(x,np);      % Calculate FFT of x
        r = ifft(xf.*conj(xf)); % Calculate biased autocorrelation
        r = real(r(1:nx))/N; % Eliminate superfluous zeros and nearly zero imaginary part
        r = [r(no:-1:2);r(1:no)]; % Make two-sided
        rh = abs(fft(r.*wn,2*NZ)); % Window autocorrelation and calculate the estimate
        Rh(c2,:) = rh(kh).';
    end;

    Rha = mean(Rh); % Average
    Bias(c1,:) = R-Rha;
    Var(c1,:) = mean((Rh - ones(NA,1)*Rha).^2);
    MSE(c1,:) = mean((Rh - ones(NA,1)*R).^2);
    drawnow;
end;

```

```

%=====
% Plot True PSD
%=====
figure;
FigureSet(1,'LTX');
subplot(2,1,1);
    h = plot(f,R,'r');
    set(h,'LineWidth',1.3);
    hold on;
        h = plot([1;1]*ef,[0;max(R)*1.2],'k');
        hold off;
    box off;
    ylabel('PSD');
    title('True PSD');
    xlim([0 pi]);
    ylim([0 max(R)*1.2]);
subplot(2,1,2);
    h = semilogy(f,R,'r');
    set(h,'LineWidth',1.3);
    hold on;
        h = plot([1;1]*ef,[0.2*min(R);max(R)*5],'k');
        hold off;
    box off;
    ylabel('PSD');
    xlim([0 pi]);
    ylim([0.2*min(R) max(R)*5]);
xlabel('Frequency (cycles/sample)');
AxisSet(8);
print -depsc BlackmanTukeyBiasVariancePSD;

```

```

%=====
% Plot Bias Variance Tradeoff
%=====
figure;
FigureSet(2, 'LTX');

for c1=1:length(ef),
    [jnk,id] = min(abs(ef(c1)-f));
    subplot(2,2,c1);
    h = plot(L,Var(:,id),L,Bias(:,id).^2,L,Var(:,id)+Bias(:,id).^2,L,MSE(:,id));
    set(h(3), 'LineWidth',3);
    set(h(4), 'LineWidth',1);
    xlim([min(L) max(L)]);
    ylim([0 1.05*max(MSE(:,id))]);
    box off;
    ylabel(sprintf('\omega = %5.3f rad/sample',ef(c1)));
    if c1==4,
        xlabel('Autocorrelation Window Length, L (samples)');
    elseif c1==1,
        title(sprintf('N:%d NA:%d',N,NA));
    elseif c1==3,
        xlabel('Autocorrelation Window Length, L (samples)');
        AxisSet(5);
        legend('Variance','Bias^2','Variance + Bias^2','MSE');
    end;
end;
AxisSet(5);
print -depsc BlackmanTukeyBiasVariance;

%=====
% Plot Bias Variance Tradeoff

```

```

%=====
figure;
FigureSet(1,'LTX');

sMSE = zeros(size(MSE));
for c1=1:length(f),
    sMSE(:,c1) = (MSE(:,c1)-min(MSE(:,c1)))/(max(MSE(:,c1))-min(MSE(:,c1)));
end;
h = imagesc(f,L,sMSE,[0 1]);
hold on;
h = plot([1;1]*ef,[0;max(L)],'k',[1;1]*ef,[0;max(L)],'w');
set(h(1:4),'LineWidth',2);
set(h(5:end),'LineWidth',1);
hold off;
[jnk,id] = min(sMSE);
hold on;
plot(f,L(id),'w. ');
hold off;
xlabel('Frequency (rad/sample)');
ylabel('Window Length (samples)');
AxisSet(8);
print -depsc BlackmanTukeyBiasVarianceBest;

```

Periodogram Averaging

$$\hat{R}_x^{(\text{PA})}(e^{j\omega}) \triangleq \frac{1}{K} \sum_{i=0}^{K-1} \hat{R}_x(e^{j\omega})$$

- Consider K IID random variables
- The variance of the mean is $1/K$ times the variance of the random variables
- If we had different independent realizations of the process we could average them and obtain a similar reduction in the variance of the PSD
- In most situations, only a single realization is available
- Can subdivide the record into K possibly overlapping segments of length L

Welch-Bartlett Spectral Estimation

$$\hat{R}_x^{(\text{WB})}(e^{j\omega_k}) \triangleq \frac{1}{K} \sum_{i=0}^{K-1} \frac{1}{L} \left| \sum_{n=0}^{L-1} v_i(n) e^{-j\omega n} \right|^2 = \frac{1}{KL} \sum_{i=0}^{K-1} |V_i(e^{j\omega})|^2$$

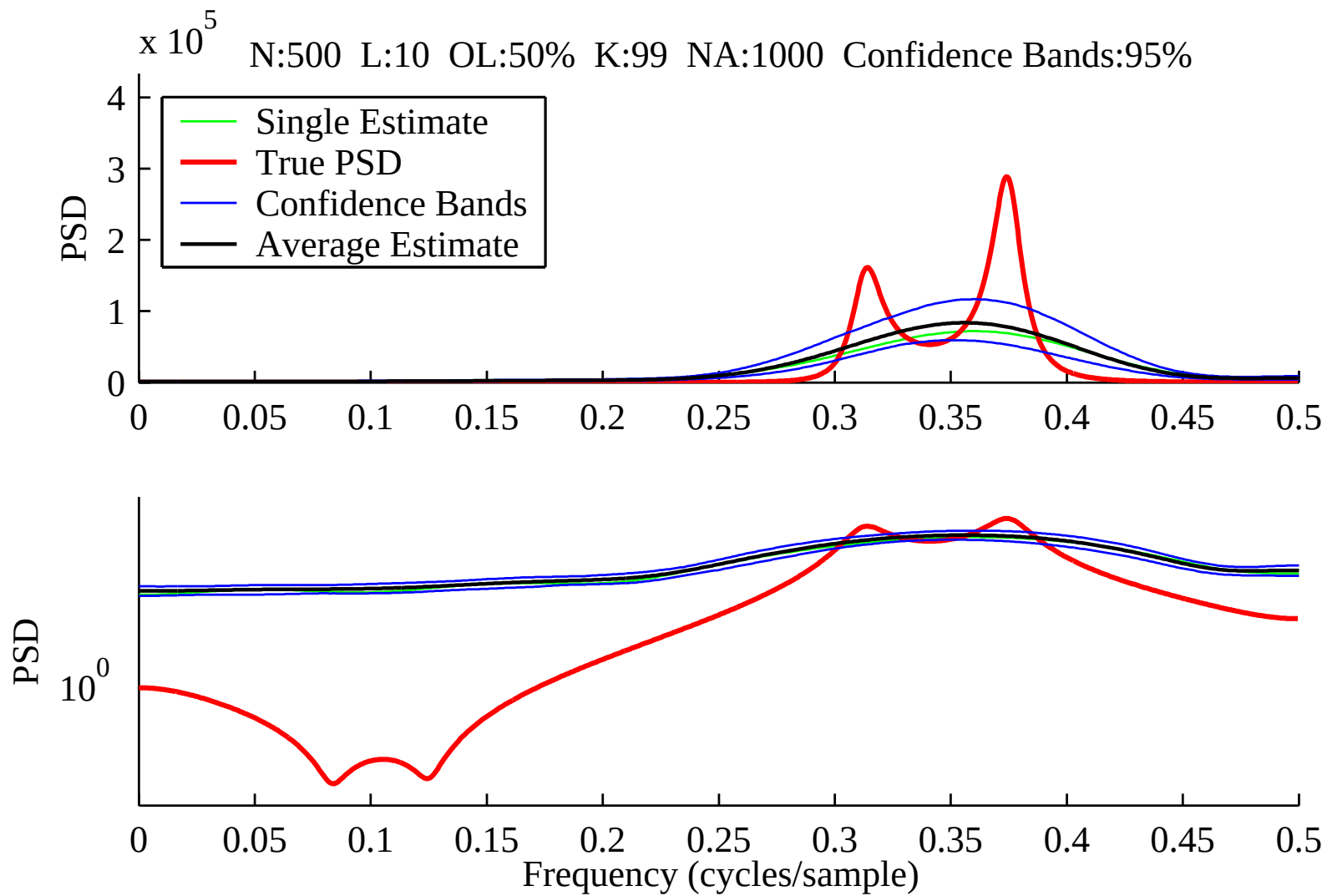
where $v(n) \triangleq x(n)w(n)$.

- Where $w(n)$ is called the **data window**
- Window does not need to be positive definite
- Window trades sidelobe leakage (ripples) for mainlobe width (blurring)
- If no overlap between windows, called the **Bartlett** estimate
- If overlap, called **Welch's** method

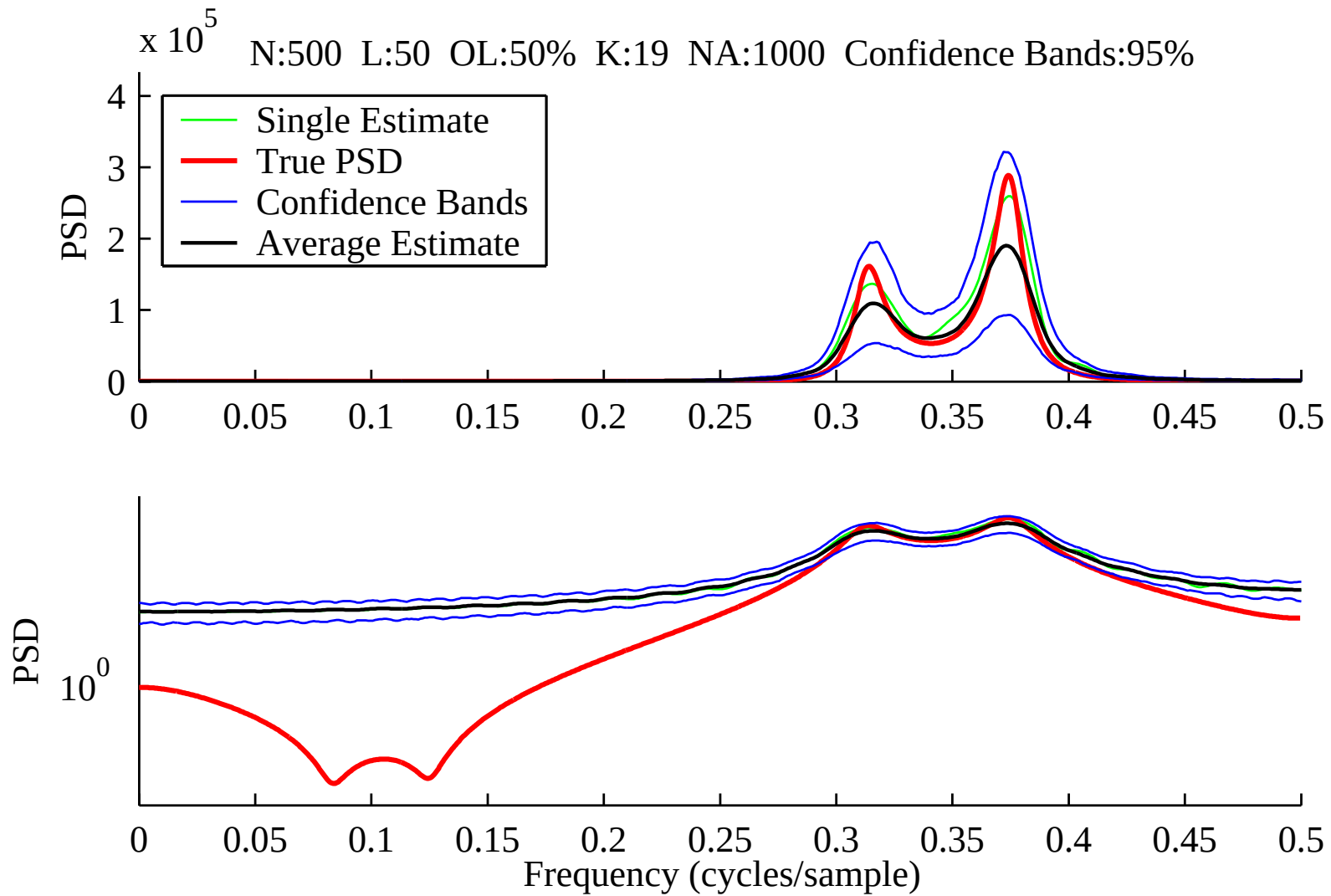
Example 13: Welch-Bartlett Spectral Estimation

Generate the Welch-Bartlett estimated PSD for a signal from a known LTI system with two sets of complex conjugate poles close to one another and near the unit circle. Repeat for various segment lengths.

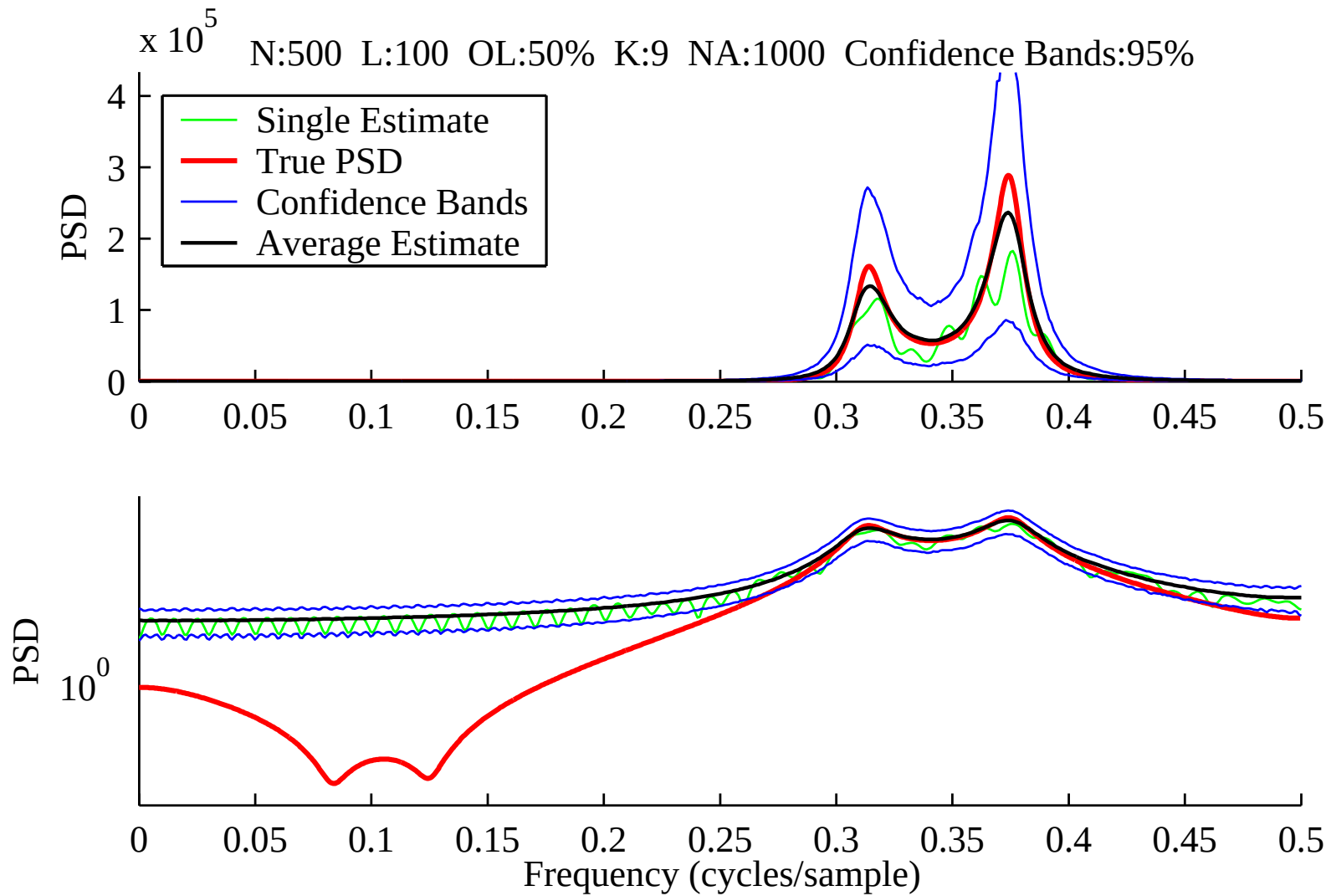
Example 13: Welch-Bartlett Estimate



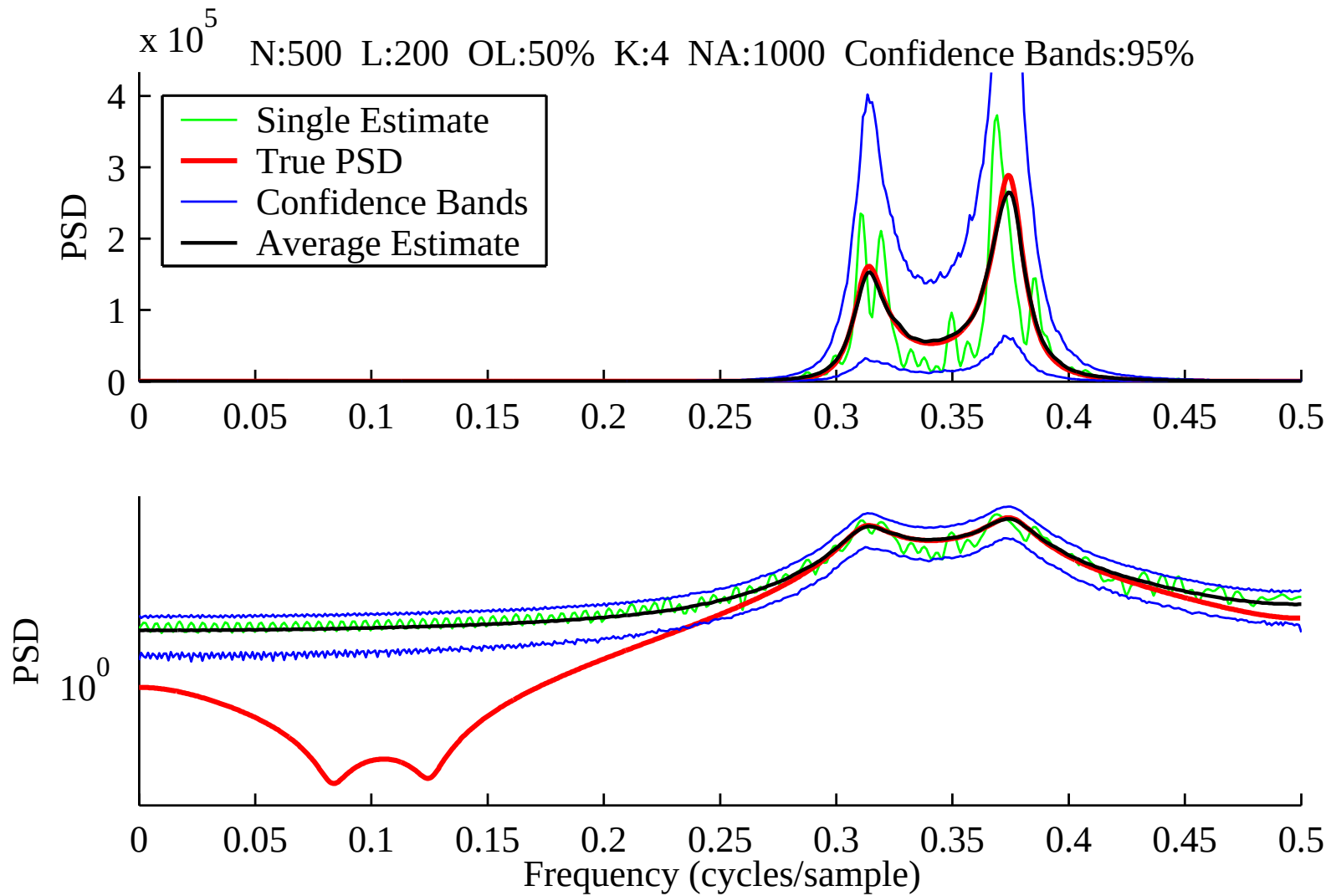
Example 13: Welch-Bartlett Estimate



Example 13: Welch-Bartlett Estimate



Example 13: Welch-Bartlett Estimate



Example 13: MATLAB Code

```
%function [] = Windows;
close all;
clear all;
L = [10,50,100,200];           % Segment lengths
O = 0.50;                      % Overlap
M = 500;                       % Padding to eliminate transient
N = 500;                       % Length of signal segment
NA = 1000;                     % No. averages
cb = 95;                       % Confidence Bands
NZ = 1024;

b = poly([-0.8,0.97*exp(j *pi/4),0.97*exp(-j *pi/4),...
          0.97*exp(j *pi/6),0.97*exp(-j *pi/6)]); % Numerator coefficients
a = poly([ 0.8,0.95*exp(j*3*pi/4),0.95*exp(-j*3*pi/4),...
          0.95*exp(j*2.5*pi/4),0.95*exp(-j*2.5*pi/4)]); % Denominator coefficients
b = b*sum(a)/sum(b);           % Scale DC gain to 1

[R,w] = freqz(b,a,NZ);
R2     = abs(R).^2;
f      = w/(2*pi);

%=====
% Plot Estimate
%=====
for c1 = 1:length(L),
    K = floor((N-O*L(c1))./(L(c1)-O*L(c1))); % No. segments
    ss = floor(L(c1)-O*L(c1));               % Step size
```

```

nx = M+N;
Rh = zeros(NA,NZ/2+1);
kh = 1:NZ/2+1;
fh = (kh-1)/NZ;
for c2 = 1:NA,
    w = randn(M+N,1);
    x = filter(b,a,w);      % System with known PSD
    x = x(nx-N+1:nx);      % Eliminate start-up transient (make stationary)

    rh = zeros(NZ,1);
    for c3 = 1:K,
        id = (1:L(c1)) + (c3-1)*ss;
        rh = rh + abs(fft(x(id),NZ)).^2;
    end;
    rh = rh/(K*L(c1));
    Rh(c2,:) = rh(kh).';
end;

Rha = mean(Rh);            % Average
Rhu = prctile(Rh,100-(100-cb)/2); % Upper confidence band
Rhl = prctile(Rh,      (100-cb)/2); % Lower confidence band

figure;
FigureSet(1,'LTX');
subplot(2,1,1);
h = plot(fh,Rh(1,:), 'g',f,R2, 'r',fh,Rhl, 'b',fh,Rhu, 'b',fh,Rha, 'k');
set(h(1), 'LineWidth',0.5);
set(h(2), 'LineWidth',1.1);
set(h(3:4), 'LineWidth',0.5);
set(h(5), 'LineWidth',0.8);

```

```

        box off;
        ylabel('PSD');
        title(sprintf('N:%d L:%d OL:%d% K:%d NA:%d Confidence Bands:%d%',N,L(c1),0*100,K,NA,c1));
        xlim([0 0.5]);
        ylim([0 max(R2)*1.5]);
        AxisSet(8);
        legend(h([1 2 4 5]),'Single Estimate','True PSD','Confidence Bands','Average Estimate',2);
subplot(2,1,2);
h = semilogy(fh,Rh(1,:), 'g',f,R2, 'r',fh,Rh1, 'b',fh,Rhu, 'b',fh,Rha, 'k');
set(h(1), 'LineWidth',0.5);
set(h(2), 'LineWidth',1.1);
set(h(3:4), 'LineWidth',0.5);
set(h(5), 'LineWidth',0.8);
box off;
ylabel('PSD');
xlabel('Frequency (cycles/sample)');
xlim([0 0.5]);
ylim([0.2*min(R2) max(R2)*5]);
AxisSet(8);
print(sprintf('WelchBartlettAverage%04d',L(c1)),'-depsc');
end;

```

Welch-Bartlett Mean

$$\begin{aligned}\mathbb{E} \left[\hat{R}_x^{(\text{WB})}(e^{j\omega_k}) \right] &= \frac{1}{K} \sum_{i=0}^{K-1} \mathbb{E} \left[\frac{1}{L} |V_i(e^{j\omega})|^2 \right] \\ &= \mathbb{E} \left[\frac{1}{L} |V_i(e^{j\omega})|^2 \right] \\ &= R_x(e^{j\omega}) * \frac{1}{L} R_w(e^{j\omega}) \\ &= \frac{1}{2\pi L} \int_{-\pi}^{\pi} R_x(e^{j\omega}) R_w(e^{j(\omega-\theta)}) d\theta\end{aligned}$$

As with the periodogram, if the window is normalized such that

$$\sum_{n=0}^{L-1} w^2(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} R_w(e^{j\omega}) d\omega = L$$

then the estimate is asymptotically unbiased

Welch-Bartlett Variance

$$\text{var}\{\hat{R}_x^{(\text{WB})}(e^{j\omega_k})\} \approx \frac{1}{K} \text{var}\{\hat{R}_x(e^{j\omega_k})\} \approx \frac{1}{K} \text{var}\{\hat{R}_x^{(\text{WB})}(e^{j\omega_k})\}$$

- Assumes segments are independent
- Not true, so estimate is optimistic
- Book fails to mention this

Welch-Bartlett Concepts

$$\hat{R}_x^{(\text{WB})}(e^{j\omega_k}) \triangleq \frac{1}{K} \sum_{i=0}^{K-1} \frac{1}{L} \left| \sum_{n=0}^{L-1} v_i(n) e^{-j\omega n} \right|^2 = \frac{1}{KL} \sum_{i=0}^{K-1} |V_i(e^{j\omega})|^2$$

- Thus, the user can once again trade variance for bias
- If $N = KL$ is fixed, then can reduce variance at the expense of increased variance by increasing K and decreasing L
- The variance is insensitive to the shape of the window
- Increasing overlap up to 50% reduces variance without substantially impacting the bias
- Further overlap increases the computational requirements, but does not help bias or variance
- Confidence intervals listed in the text

Welch-Bartlett Comments

$$\hat{R}_x^{(\text{WB})}(e^{j\omega_k}) = \frac{1}{KL} \sum_{i=0}^{K-1} |V_i(e^{j\omega})|^2$$

- Consider the case when the overlap is $L - 1$ samples
- This would reduce variance the most
- In this case it becomes equivalent to periodogram smoothing [?, pp. 583–585]!
- Bartlett originally proposed averaging periodograms of separate segments, but showed that this was essentially equivalent to the Blackman-Tukey method with a “Bartlett” autocorrelation window [?, pp. 439-440]
- When the overlap is less than $L - 1$
 - It is only approximately equivalent to periodogram smoothing
 - It requires L FFTs
 - Variance is increased

Summary

- We discussed several different nonparametric methods of spectral estimation
- The “natural” estimate was the periodogram
- This estimate has unacceptable variance for most applications
- There are two fundamental approaches to reduce variance
 - Periodogram smoothing
 - Periodogram averaging
- They have comparable performance and ultimately are approximately equivalent
- Each enables the user to control the bias-variance tradeoff
- Typically the smoothing parameter is chosen to minimize bias
- Only approximate confidence intervals are available for each

MATLAB Resources

- The MATLAB implementation of these functions is constantly changing (improving?)
- The documentation is pretty good
 - Open up the help window (helpwin)
 - Signal Processing Toolbox → Statistical Signal Processing
- Most of the techniques we have discussed are implemented
- Interface is not user-friendly

References